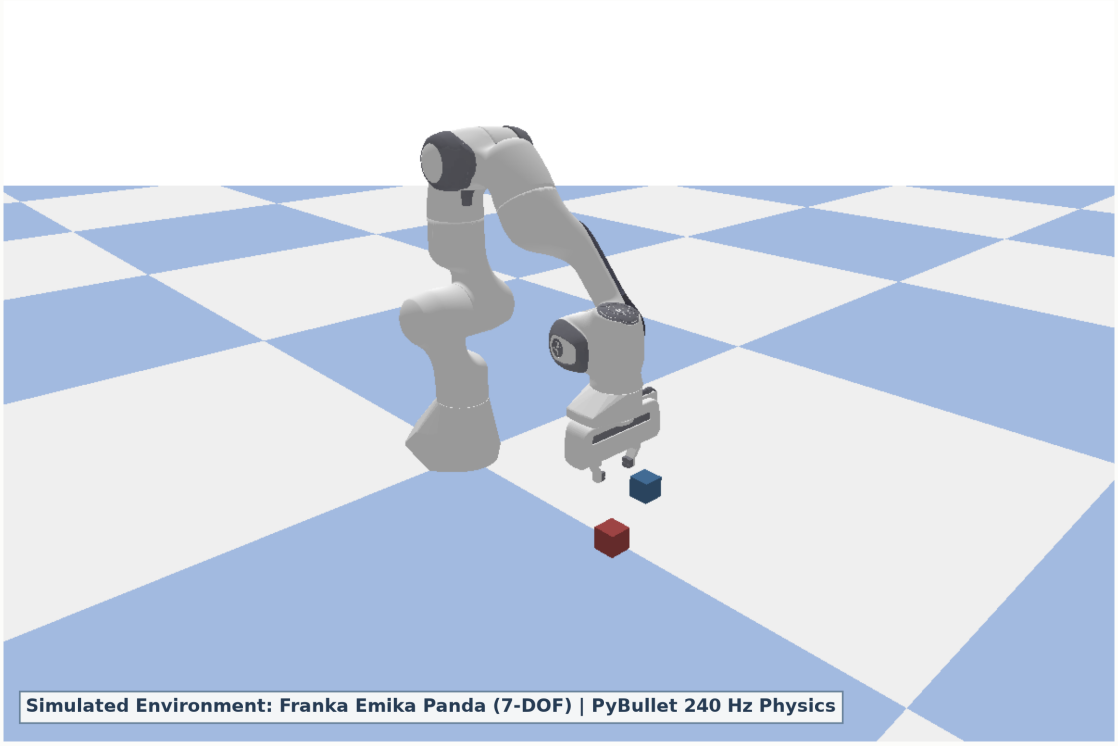
TECHNICAL RESEARCH MONOGRAPH | PROJECT: DEXTEROUS-ROBOT-LEARNING

DEXTEROUS ROBOTIC LEARNING

Adaptive Manipulation, Reinforcement Learning, Robust Control and Simulation-Based Evaluation



Simulated Environment: Franka Emika Panda (7-DOF) | PyBullet 240 Hz Physics

Monograph Attribute	Verified Specification
Project	Dexterous-Robot-Learning (Franka Emika Panda 7-DOF Platform)
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Document Version & Date	Monograph v17.0.0 — Released 2026-11-12
Verification Scope	Phases 1–17 Simulation-Verified (PyBullet & MuJoCo 4.0.1); Phase 17.3 Physical Robot: NOT_ATTEMPTED; Phase 18: Proposed Future Scope
Regression & Software Baseline	126/126 Combined Python Tests (PyBullet + FastAPI/WS) | TypeScript + Vite Build + Vitest Pass

Document Metadata, Licensing Notice, Disclaimer, and Abstract

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> SCIENTIFIC DISCLAIMER: All reinforcement learning policies, adaptive controllers, vision unprojection models, stress benchmarks, and E-Stop timing bounds documented in Phases 1–17 were executed exclusively in CPU-only physics simulators (PyBullet and MuJoCo 4.0.1). Physical Franka Emika Panda hardware control (Phase 17.3) was NOT_ATTEMPTED. No peer-reviewed acceptance or physical factory deployment is claimed.

Preface

This monograph transforms seventeen incremental phases of robotics research and software refactoring into an integrated educational and engineering reference. Rather than masking negative outcomes or skipping foundational mathematics, each algorithmic decision—from Damped Least-Squares kinematics and Coulomb contact friction cones to hindsight goal relabeling failures and sim-to-sim falsification—is derived from first principles and traced verbatim to source code and experimental logs.

Abstract

Dexterous manipulation with redundant 7-DOF arms requires synchronized stability across inverse kinematics, bilateral contact mechanics, reward geometry, and state estimation. Across Phases 1–17 of Dexterous-Robot-Learning, we investigate SAC+HER sparse deadlocks (25x100k runs = 0.0% due to static-goal hindsight relabeling), achieve 100.0% dense Pick-and-Place convergence at 600k–1M steps (0.0142 m precision), diagnose Stage-3 OOD slippage (68.4 N -> 0.0 N force collapse), and boost held-out DR4 generalization from 57.0% to 98.0% (+41.0 pp) via zero-retraining runtime shields (Controller E). Subsequent extensions validate 50/50 (100%) two-object PreGrasp sorting, 75% grasp / 45% lift with progress-masked Behavioral Cloning, 30.0% HRL sample efficiency at 12k steps vs. 0% flat SAC, 14.8 mm oblique RGB-D unprojection, 95–100% Kalman dynamic interception, 53.33% compliant peg-in-slot assembly, MuJoCo 4.0.1 sim-to-sim gap quantification, and an authoritative FastAPI + Three.js digital twin workstation verified by 126 passing automated Python tests.

Keywords: Dexterous Manipulation, Franka Emika Panda, Soft Actor-Critic, Hindsight Experience Replay, Behavioral Cloning, Hierarchical Reinforcement Learning, Domain Randomization, Coulomb Friction, Sim-to-Sim Transfer, Robotics Digital Twin.

Table of Contents, Figures, Tables, and Technical Notation

Contents (Clickable Chapter & Section Index)

Chapter / Section	Page
Chapter 1: Robotic Manipulation, Project Objectives, and Historical Scope	p. 4
Chapter 2: Kinematics, Jacobian Control, Contact Mechanics, and Coulomb Friction	p. 7
Chapter 3: Reinforcement Learning, SAC, HER, Imitation, Randomization, and HRL	p. 10
Chapter 4: Modular System Stack, Simulation Design, and Engineering Verification	p. 14
Chapter 5: Early Foundations Through Sparse Deadlocks (Phases 1–11)	p. 16
Chapter 6: 1M Breakthrough, Adaptive Control, and Multi-Task Scaling (Phases 12–15)	p. 18
Chapter 7: HRL, Oblique RGB-D Vision, Dynamic Tracking & MuJoCo Parity (Phases 16–17)	p. 22
Chapter 8: Cross-Phase Synthesis, Negative Results, and Provenance Analysis	p. 25
Chapter 9: Refactored Robotics Workstation, FastAPI Core, and Three.js Digital Twin	p. 27
Chapter 10: Future Scope — Multi-Agent Cooperative Manipulation Architecture	p. 29
Chapter 11: Verified Conclusions, Bibliography, Further Reading, and Registries	p. 31

List of Figures & Technical Illustrations

Figure Caption	Page
Figure 1.1: Real-world laboratory compliant 7-DOF manipulator grasping a machined comp...	p. 5
Figure 1.2: Franka Emika Panda 7-DOF kinematic chain, effort ratings, joint limits, an...	p. 5
Figure 2.1: Jacobian inversion, damping lambda=0.05 conditioning, and null-space postu...	p. 7
Figure 2.2: (a) Bilateral gripper free-body force diagram; (b) Minimum required normal...	p. 8
Figure 2.3: Real-world parallel-jaw robotic gripper and wrist vision camera in a unive...	p. 9
Figure 3.1: Off-policy SAC actor-critic dataflow coupled with HER k=4 virtual relabeli...	p. 11
Figure 3.2: (a) Sparse plateau vs. dense gradient; (b) Solving the z=0.50 m hover stal...	p. 12
Figure 3.3: Transition from unbalanced closed-jaw BC bias (5B1) to balanced masked BC ...	p. 12
Figure 3.4: Calibrated physical mass and friction parameter spans across Domain Random...	p. 13
Figure 4.1: Complete four-layer Dexterous-Robot-Learning architecture unifying physics...	p. 14
Figure 4.2: Authoritative PyBullet 240 Hz stepping cycle, adaptive lift guard, DLS IK ...	p. 14
Figure 5.1: Reconstructed Phases 2–8 TensorBoard actor loss and entropy coefficient de...	p. 16
Figure 5.2: Phase 11 25-model sparse stagnation across 5,000 evaluation episodes (Mean...	p. 17
Figure 6.1: Phase 12 1M-step dense Pick-and-Place S-curve (N=100 evaluation episodes p...	p. 18
Figure 6.2: Phase 13 Matched vs. Held-Out DR4 generalization gap (multi-seed Mean +/- ...	p. 18
Figure 6.3: Real Phase 13.5 D5 telemetry contrasting successful lift (ep5001) against ...	p. 19
Figure 6.4: Phase 14 Adaptive Manipulation Shield integrated with the Phase 15.2.5A fi...	p. 19
Figure 6.5: Phase 14.4 Held-Out DR4 confirmation across Controllers A, D, and E (Seeds...	p. 20
Figure 6.6: Sequential dual-cube MultiSort workflow (50/50 scripted) and SE(2)+Z compl...	p. 20
Figure 6.7: Phase 15.1 stress resilience, 15.2.5B–B8 grasp/lift ablations, and Phase 1...	p. 21
Figure 7.1: Integration across oblique RGB-D perception, 5-skill HRL selection, Kalman...	p. 22
Figure 7.2: Empirical Phase 16 suite across HRL (12k), RGB-D Vision (20 ep), Dynamic K...	p. 23
Figure 7.3: Live PyBullet 3D oblique camera frustum (FOV 55 deg) and unprojected 91-pi...	p. 23
Figure 7.4: Phase 17.1 PyBullet-to-MuJoCo 4.0.1 transfer gap and Phase 17.2 simulated ...	p. 24
Figure 9.1: Authoritative FastAPI SimulationManager, REST validation gates, 20 Hz WebS...	p. 27
Figure 9.2: Three-dimensional workspace bounds and multi-target Panda kinematic chains...	p. 27
Figure 10.1: Proposed Phase 18 Centralized Training / Decentralized Execution (CTDE) M...	p. 30
Figure 10.2: Multi-axis industrial serial manipulator illustrating industrial automati...	p. 30

List of Quantitative & Reference Tables

Table Caption	Page
Table 4.1: Verified Gymnasium & Cross-Simulator Environment Contracts	p. 15
Table 8.1: Cross-Phase Defect Root-Cause Registry (ERR-001 through ERR-012)	p. 26
Table 11.1: Curated Robotics & Reinforcement Learning Study Guide	p. 32
Table 11.2: Representative Executed Experiments Ledger (Strict Provenance Separation)	p. 33
Table 11.3: Phase 0–17 Verification Status Summary	p. 35

List of Abbreviations & Mathematical Symbols

Symbol / Acronym	Definition
7-DOF / SE(3)	Seven Degrees of Freedom / Special Euclidean 3D rigid-body pose group
DLS IK / J(q)	Damped Least-Squares Inverse Kinematics / Manipulator Jacobian R^(6x7)
SAC / HER	Soft Actor-Critic (Auto-Entropy alpha) / Hindsight Experience Replay (k=4)
BC / Hybrid	Behavioral Cloning (masked M_prog) / Decaying Auxiliary BC-SAC
HRL / SMDP	Hierarchical Reinforcement Learning / Semi-Markov Decision Process Options
DR0 .. DR4	Domain Randomization sensitivity tiers (nominal to held-out OOD extremes)
F_n / mu	Bilateral finger normal squeeze force (N) / Coulomb lateral friction coeff
GBNL	GRASP_BUT_NO_LIFT failure mode under impulsive Stage-3 vertical acceleration
FCI / HAL	Franka Control Interface / Hardware Abstraction Layer (Spec Only; NOT_ATTEMPTED)

PART I — FOUNDATIONS

Chapter 1: Robotic Manipulation, Project Objectives, and Historical Scope

1.1 The Fundamental Challenge of Contact-Rich Manipulation

Autonomous robotic manipulation differs profoundly from unconstrained motion planning or classical game benchmarks. When a 7-degree-of-freedom (7-DOF) serial manipulator approaches, grasps, lifts, transports, and inserts rigid objects in a tabletop workspace, control policies must negotiate discontinuous contact transitions, frictional shear limits across millimeter jaw apertures, non-convex kinematic joint bounds, partial observation noise, and severe long-horizon credit-assignment delays. A millimeter lateral misalignment or an excessive 0.14 m/s^2 vertical acceleration pulse at grasp closure can instantly break bilateral friction equilibrium and drop a payload.

The Dexterous-Robot-Learning project investigates how off-policy reinforcement learning (Soft Actor-Critic paired with Hindsight Experience Replay), dense potential shaping, progress-masked Behavioral Cloning, Hierarchical Reinforcement Learning (HRL), oblique RGB-D computer vision, and physics-informed inference shields can be systematically composed to achieve reliable simulated manipulation on the Franka Emika Panda platform under strict CPU-only single-process compute constraints.

1.2 Verified Research Scope (Phases 1–17 Active; Phase 18 Proposed)

Project development spans Phases 0 through 17 in verifiable simulation artifacts (PyBullet primary physics engine and MuJoCo 4.0.1 cross-simulator verification), alongside a refactored web-based simulation workstation built over FastAPI and Three.js. Phases 1–8 are reconstructed strictly from Git history (earliest commits 3a345f3, a5cc06b, 7ad9848 dated 2026-10-03), YAML snapshots, and TensorBoard scalars; where standalone historical artifacts (such as standalone Phase 5 notes) do not exist in the repository, they are explicitly documented as NOT RECOVERABLE FROM AVAILABLE EVIDENCE. Phase 17.3 hardware execution on physical Franka hardware was recorded as NOT_ATTEMPTED, and Phase 18 (Multi-Agent Cooperative Manipulation) represents proposed future research scope only.

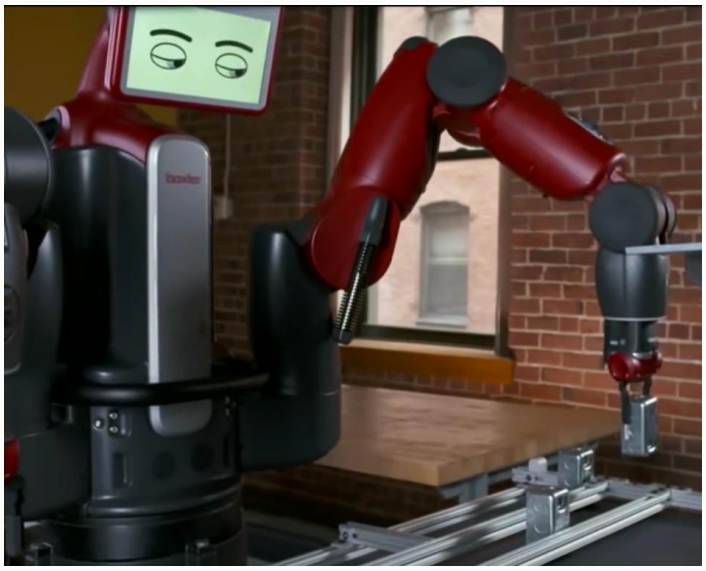


Figure 1.1: Real-world laboratory compliant 7-DOF manipulator grasping a machined component above an industrial test rail. Source: Wikimedia Commons (File:Baxter_1.jpg), Author: Salammbo31, License: CC BY-SA 4.0. Note: Illustrates general laboratory manipulation context; the current project evaluates Franka Panda strictly in simulation.

1.3 Franka Emika Panda Architecture and PyBullet Simulation Fundamentals

In real industrial and university laboratories, torque-controlled 7-DOF collaborative manipulators such as the Franka Emika Panda provide kinematic redundancy: seven revolute axes allow reaching an arbitrary 6-DOF end-effector pose (position [x, y, z] and orientation [qx, qy, qz, qw]) while exploiting the extra null-space degree of freedom to avoid joint stops and singular elbow locking. Within simulation (`src/robots/panda.py`, wrapped cleanly in `robotics/robots/franka_panda/robot_model.py`), the official URDF `panda.urdf` models arm joints [0..6] with maximum effort limits [87, 87, 87, 87, 12, 12, 12] N*m, finger joints [9, 10] with 25 N maximum squeeze force over [0.00, 0.08] m aperture, and grasp target frame link 11 (`panda_grasptarget`).

Franka Emika Panda 7-DOF Kinematic Chain, Joint Limits, Torque Ratings, and Link Transforms

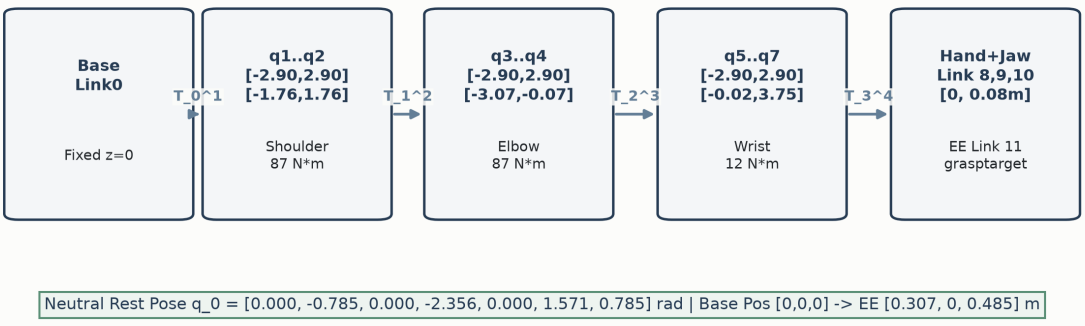


Figure 1.2: Franka Emika Panda 7-DOF kinematic chain, effort ratings, joint limits, and neutral reference posture q_0.

1.4 Real-World Applications and Engineering Relevance

Manipulator architectures of this class underpin modern flexible automation across: (1) Industrial bin picking and machine tending (extracting randomized parts from shallow trays); (2) Warehouse order fulfillment and parcel sorting (rapid sequential color/SKU binning analogous to Phase 15.2 MultiSort); (3) Precision manufacturing assembly (connector mating and peg-in-slot insertion under lateral wall forces analogous to Phases 15.3 and 16.4); and (4) Healthcare and assistive robotics (adaptive object transfer near fragile surfaces). Deploying simulated policies onto hardware in any of these sectors would require empirical system identification, sensor calibration, real-time FCI driver certification (1 kHz control loop), and hardware safety interlocks beyond current simulation scope.

PART II — MATHEMATICAL & ALGORITHMIC FOUNDATIONS

Chapter 2: Kinematics, Jacobian Control, Contact Mechanics, and Coulomb Friction

2.1 Forward Kinematics, Inverse Kinematics, and Jacobian Control

A. Definition & Purpose: Forward Kinematics (FK) maps joint coordinates q in R^7 into end-effector Cartesian pose x = FK(q) in SE(3) through homogeneous link transforms T_0^1(q_1)...T_6^7(q_7). Conversely, Inverse Kinematics (IK) determines joint increments dq achieving a desired Cartesian displacement dx, solving the problem of commanding intuitive 3D task goals [dx, dy, dz] with a 7-DOF arm.

B. Why Selected & Mathematical Formulation: Rather than asking RL policies to directly regress 7 coupled joint torques (which requires orders of magnitude more exploration samples), this repository exposes compact 4D/5D task-space actions mapped via PyBullet's Damped Least-Squares (DLS) Jacobian pseudoinverse (Equation 1–2). The Jacobian J(q) in R^(6x7) relates joint velocities dq/dt to spatial twist dx/dt = J(q) dq/dt. Pure pseudoinverses J^T (J J^T)^(-1) explode near kinematic singularities where det(J J^T) -> 0; adding damping lambda = 0.05 stabilizes inversion while projecting neutral posture regularization (I - J# J) k_0 (q_0 - q_t) into the 1-DOF null space.

$$\mathbf{q}_{t+1} = \mathbf{q}_t + J^\top (JJ^\top + \lambda^2 I)^{-1} \Delta \mathbf{x} + (I - J^{\#} J) k_0 (\mathbf{q}_0 - \mathbf{q}_t) \quad \textbf{(1)}$$

Equation (1): Damped Least-Squares Inverse Kinematics with null-space posture regularization.

$$\dot{\mathbf{x}} = J(\mathbf{q})\dot{\mathbf{q}}, \quad J(\mathbf{q}) \in \mathbb{R}^{6\times 7} \quad \textbf{(2)}$$

Equation (2): Instantaneous Jacobian velocity mapping for 7-DOF spatial twist.

Redundant 7-DOF Jacobian Inversion, Damping Stabilization, and Null-Space Posture Projection

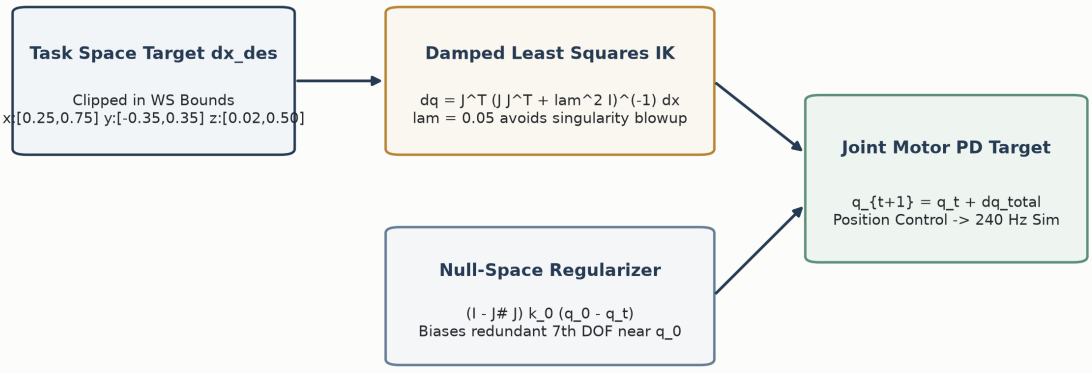


Figure 2.1: Jacobian inversion, damping lambda=0.05 conditioning, and null-space posture regularization in FrankaPanda.calculate_ik.

Code Excerpt — src/robots/panda.py :: FrankaPanda.calculate_ik & clip_to_workspace

```
clipped_pos = self.clip_to_workspace(target_position)
all_ik_angles = p.calculateInverseKinematics(
    bodyUniqueId=self.body_id, endEffectorLinkIndex=self.ee_link_idx,
    targetPosition=list(clipped_pos), targetOrientation=orn,
    lowerLimits=self.joint_lower_limits, upperLimits=self.joint_upper_limits,
    jointRanges=self.joint_ranges, restPoses=self.neutral_joint_positions,
    maxNumIterations=100, residualThreshold=1e-4, physicsClientId=self.cid,
)
# Clips XYZ within ws_x=[0.25,0.75], ws_y=[-0.35,0.35], ws_z=[0.02,0.50]; solves DLS IK in <1ms.
```

C. Practical Example, Alternatives & Real-World Context: In Phase 1, 5 validation waypoints confirmed 0.7 mm to 4.5 mm tracking accuracy. Alternatives include QP whole-body velocity controllers (OSQP) and operational-space torque control, which offer superior dynamic impact compliance at the cost of requiring accurate inertia matrices. In real manufacturing and surgical arms, damped Jacobian IK forms the universal inner kinematics layer beneath high-level planners.

2.2 Contact Mechanics, Parallel-Jaw Grasping, and Coulomb Friction Cones

A. Definition & Internal Physics: Grasp stability during lifting is governed by unilateral normal squeeze forces F_{n,9}, F_{n,10} applied by Panda finger links 9 and 10 onto opposite cube faces, creating shear load capacity limited by Coulomb's friction cone |f_s| <= mu * F_n (Equation 9). To elevate an object of mass m under upward liftoff acceleration a_z without slip, total bilateral squeeze force must obey F_{n,req} >= m(g + a_z) / (2 * mu).

$$|f_s| \le \mu F_n, \quad F_{n,\mathrm{req}} \ge \frac{m(g+a_z)}{2\mu} \tag{9}$$

Equation (9): Coulomb friction cone limit and minimum normal squeeze force invariant.

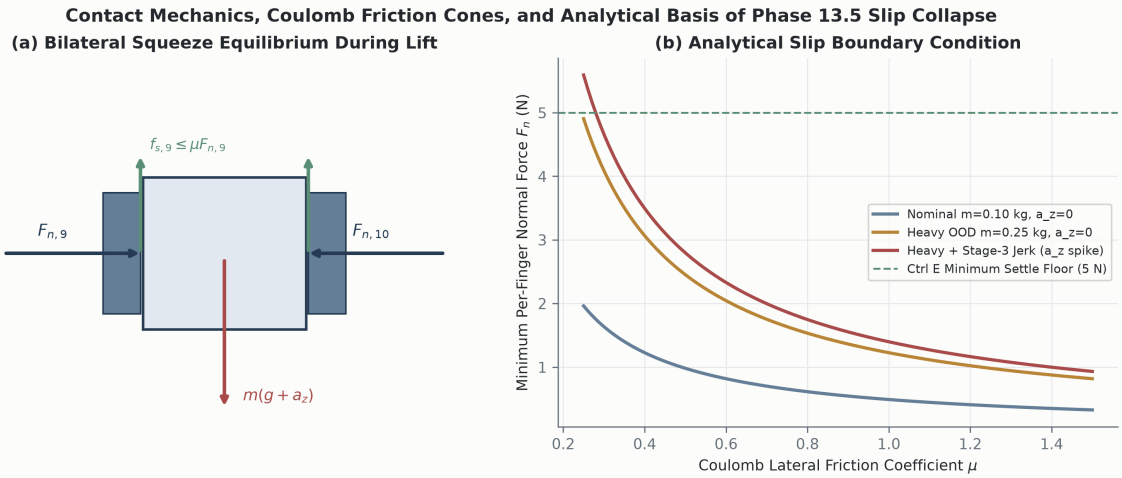


Figure 2.2: (a) Bilateral gripper free-body force diagram; (b) Minimum required normal force across friction mu showing OOD slip cliff.

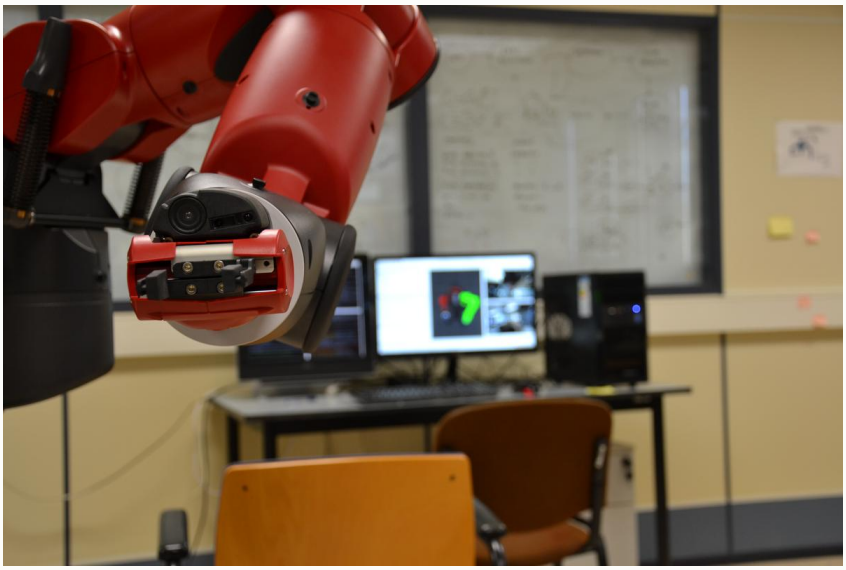


Figure 2.3: Real-world parallel-jaw robotic gripper and wrist vision camera in a university research workstation. Source: Wikimedia Commons (File:Baxter_robot's_gripper.JPG), Author: Gustavo A. Casan, License: CC BY-SA 4.0.

B. Evidence & Project Lessons Learned: Two major failures traced directly back to Equation (9). First, in Phase 6 (ERR-001), default PyBullet finger friction could not hold a 0.10 kg cube; configuring lateralFriction=2.0 and spinningFriction=0.1 fixed scripted pick-and-place. Second, in Phase 13.5, heavy OOD cubes (0.25 kg) under low friction (mu=0.3) combined with an impulsive a_z=0.14 m/s^2 liftoff jerk spiked normal force to 68.4 N before snapping contact to 0.0 N (30.5% GRASP_BUT_NO_LIFT failure rate), directly inspiring Phase 14 Controller E's 3-tick settle + lift-rate safety shield.

PART II — MATHEMATICAL & ALGORITHMIC FOUNDATIONS

Chapter 3: Reinforcement Learning, SAC, HER, Imitation, Randomization, and HRL

3.1 Goal-Conditioned Markov Decision Processes and Soft Actor-Critic (SAC)

A. Definition & Role: Task execution is modeled as a Goal-Conditioned MDP (S, A, G, P, r, gamma), where observations follow the Gymnasium GoalEnv Dict schema {observation, achieved_goal, desired_goal}. Soft Actor-Critic (SAC) is an off-policy maximum-entropy actor-critic algorithm that simultaneously maximizes expected discounted return and expected policy entropy H(pi(.|s,g)).

B. Internal Mechanics & Equations: SAC trains twin Q-critics Q_{theta_1}, Q_{theta_2} against Polyak-averaged targets (tau=0.05 early; 0.005 late) using clipped double-Q minimization to curb overestimation bias (Equations 3–4). The stochastic actor samples continuous 4D actions via the reparameterization trick a = tanh(mu_phi(s,g) + sigma_phi(s,g)*eps) and optimizes Equation (5), while automatic Lagrange dual updates adapt temperature alpha toward target entropy H_tgt = -dim(A) = -4 (Equation 6).

$$\mathcal{L}_Q(\theta_i) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}, g) \sim \mathcal{D}}\left[\left(Q_{\theta_i}(s_t, a_t, g) - y\right)^2\right] \tag{3}$$

Equation (3): Clipped double Q-learning critic regression objective.

$$y = r_t + \gamma\left(\min_{i=1,2} Q_{\bar{\theta}_i}(s_{t+1}, \tilde{a}_{t+1}, g) - \alpha \log \pi_\phi(\tilde{a}_{t+1} \mid s_{t+1}, g)\right) \tag{4}$$

Equation (4): Soft Bellman backup target with entropy bonus.

$$J_\pi(\phi) = \mathbb{E}_{s_t \sim \mathcal{D}, \epsilon \sim \mathcal{N}}\left[\alpha \log \pi_\phi(f_\phi(\epsilon; s_t, g) \mid s_t, g) - \min_{i=1,2} Q_{\theta_i}(s_t, f_\phi, g)\right] \tag{5}$$

Equation (5): Reparameterized SAC actor objective.

$$\alpha^* = \arg\min_{\alpha > 0} \; \mathbb{E}_{a_t \sim \pi_\phi}[-\alpha(\log \pi_\phi(a_t \mid s_t, g) + \mathcal{H}_{\text{tgt}})], \;\; \mathcal{H}_{\text{tgt}} = -4 \tag{6}$$

Equation (6): Automatic entropy temperature dual minimization.

C. Alternatives, Limitations & Evidence: On-policy methods like PPO discard off-policy replay transitions and lack native HER compatibility; deterministic off-policy methods like DDPG/TD3 require external exploration noise tuning. Across 10k sparse runs in Phases 2–8, SAC suffered premature alpha collapse (0.907 -> 0.002), demonstrating that without informative positive gradients, entropy scheduling alone cannot bridge long manipulation horizons.

3.2 Hindsight Experience Replay (HER) and Sparse vs. Dense Reward Shaping

A. Mechanism & Static-Goal Exploration Trap (ERR-003): Under binary sparse rewards r = -I(||p_obj - p_goal|| > 0.05 m) (Equation 7), random policies almost never hit desired goals. Future-strategy HER (k=4, Equation 8) retrospectively substitutes achieved states g' = ag_tau (tau > t) into replay mini-batches, turning failed trajectories into valid goal-reaching examples. However, Phase 11 (25 models x 100k steps = 0.0% across 5,000 evaluation episodes) uncovered a critical failure mode: when an agent fails even to move the object off the table, every future achieved goal g' equals the static initial resting coordinate, reinforcing stationary hovering.

$$r_{\text{sparse}}(s_{t+1}, g) = -\mathbb{1}(\|\mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{goal}}\|_2 > \varepsilon),\ \varepsilon = 0.05\,\text{m} \tag{7}$$

Equation (7): Binary sparse Euclidean tolerance reward.

$$g' \sim \{\phi(s_\tau) : \tau > t,\ k = 4 \text{ future HER sampling}\} \tag{8}$$

Equation (8): Future-strategy HER virtual goal relabeling (k=4).

Soft Actor-Critic (SAC) Coupled with Future-Strategy HER and Static-Goal Exploration Mitigation

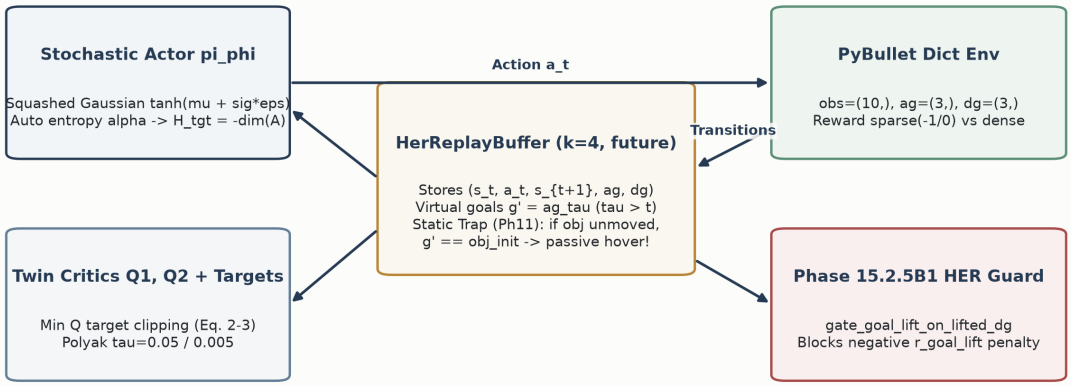


Figure 3.1: Off-policy SAC actor-critic dataflow coupled with HER k=4 virtual relabeling and the Phase 15.2.5B1 lift-gradient guard.

B. Dense Staged Shaping & Continuous Vertical Potentials: Breaking the static trap in Phase 12 required staged dense shaping (Equation 10) uniting approach attraction, bilateral contact bonuses, lift altitude increments, goal transport potential, and collision penalties. Similarly, Phase 15.2.5B1 overcame an XY-gated vertical hover deadlock at z=0.50 m by replacing cylinder-gated Z rewards with an un-gated continuous vertical potential Phi_z = -2.0 * clip(|z_ee - z_grasp|, 0, 1) (Equation 11).

$$r_{\text{dense}} = -w_r d_{\text{ee, obj}} + w_g \mathbb{1}_{\text{bilat}} + w_l \Delta z_{\text{obj}} - w_t d_{\text{obj, goal}} + B_{\text{succ}} \mathbb{1}_{\text{succ}} - w_c \mathbb{1}_{\text{coll}} \tag{10}$$

Equation (10): Multi-stage dense manipulation shaping function (Phase 12).

$$r_{\text{pregrasp}} = w_p(\Phi_z(s_{t+1}) - \Phi_z(s_t)) + r_{\text{grasp}} - w_{\text{coll}} \mathbb{1}_{\text{table}},\ \Phi_z = -2\,\text{clip}(|z_{\text{ee}} - z_g|, 0, 1) \tag{11}$$

Equation (11): Continuous un-gated vertical approach potential Phi_z (Phase 15.2.5B1).

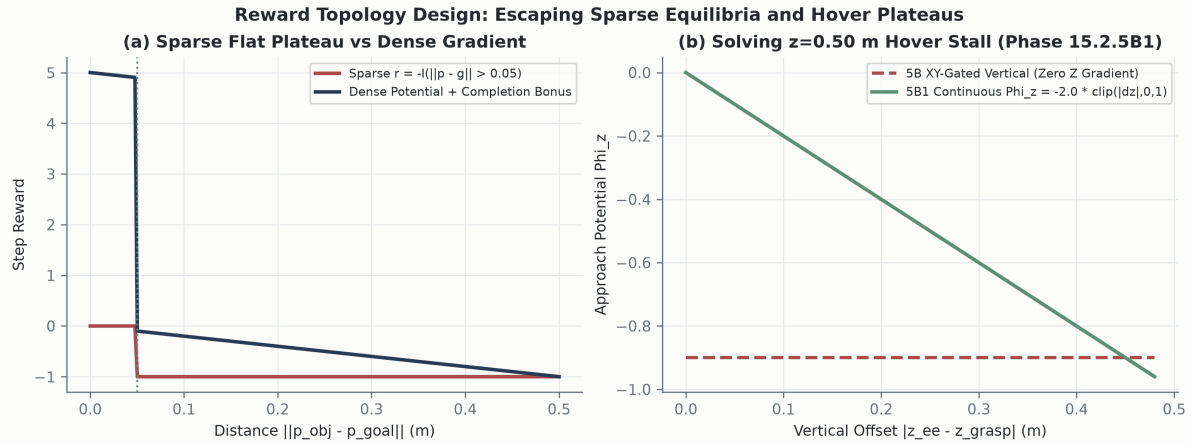


Figure 3.2: (a) Sparse plateau vs. dense gradient; (b) Solving the z=0.50 m hover stall via un-gated vertical potential Phi_z.

3.3 Progress-Masked Behavioral Cloning, Hybrid BC-SAC, Domain Randomization & HRL

Behavioral Cloning (BC) initializes actor policies from expert demonstrations via supervised regression. Phase 15.2.5B.1 proved that naive BC over complete scripted episodes overfits trailing stationary HOLD_AT_GOAL frames (78.4% closed-jaw bias), causing premature air-pinching. Introducing balanced open-approach demos (70.1% open descent) and binary progress masking M_prog(s) (Equations 12–13) propelled pure masked BC (5B7) to 75% grasp and 45% lift across 100 episodes with 0% table collisions, while Decaying Hybrid BC-SAC (5B8) reached 66% / 26%.

$$\mathcal{L}_{\mathrm{BC}}(\phi) = \mathbb{E}_{(s,\,a^{*})\sim\mathcal{D}_{\mathrm{bal}}}\left[M_{\mathrm{prog}}(s)\,\|\mu_{\phi}(s) - a^{*}\|_{2}^{2}\right] \tag{12}$$

Equation (12): Progress-masked Behavioral Cloning L2 objective.

$$J_{\mathrm{hybrid}}(\phi) = J_{\pi}^{\mathrm{SAC}}(\phi) + \lambda_{\mathrm{BC}}(t)\,\mathcal{L}_{\mathrm{BC}}(\phi), \quad \lambda_{\mathrm{BC}}(t) = \lambda_{0}e^{-\kappa t} \tag{13}$$

Equation (13): Exponentially decaying Hybrid BC-SAC objective with entropy floor.

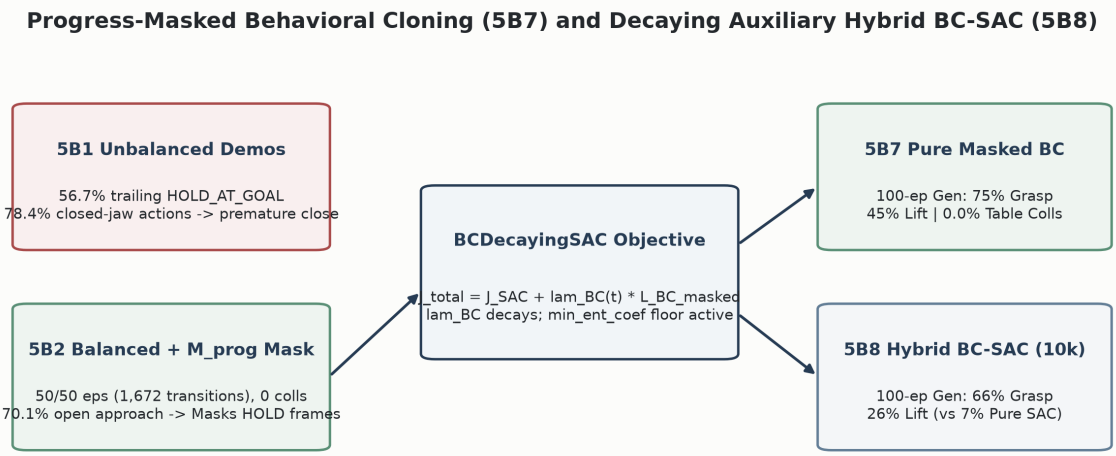


Figure 3.3: Transition from unbalanced closed-jaw BC bias (5B1) to balanced masked BC (5B7) and decaying Hybrid BC-SAC (5B8).

Domain Randomization (DR0–DR4, Figure 3.4) varies cube mass (0.04–0.25 kg), Coulomb friction (0.3–1.5), and spawn/goal geometry so policies do not overfit fixed nominal constants. Hierarchical RL (Phase 16.1,

Equation 14) further mitigates horizon complexity by training a high-level discrete meta-policy pi_high(u|s) over five modular closed-loop skill policies {REACH, ALIGN, GRASP, LIFT, TRANSPORT}.

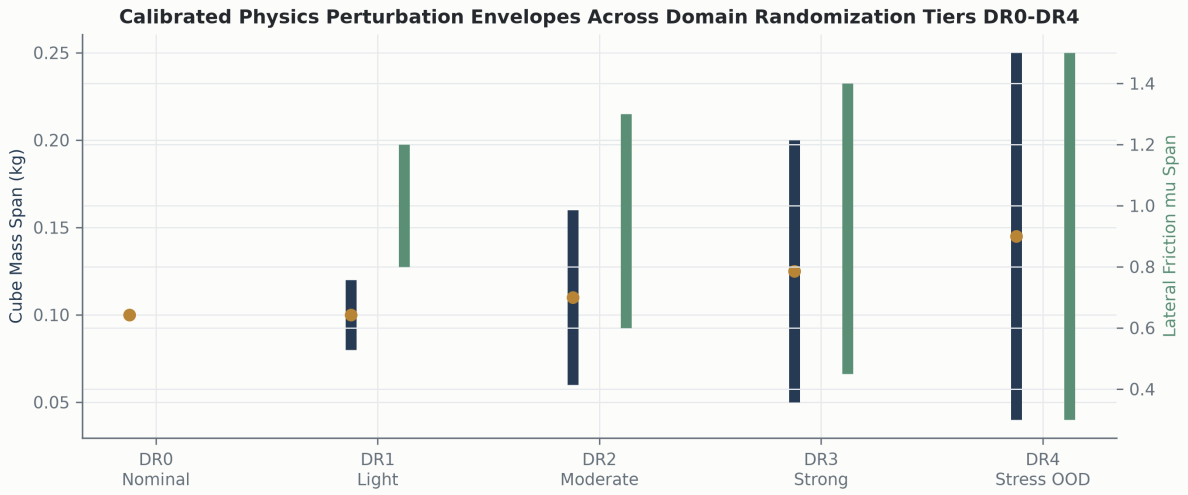


Figure 3.4: Calibrated physical mass and friction parameter spans across Domain Randomization tiers DR0 through DR4.

$$\pi_{\mathrm{HRL}}(a_t \mid s_t) = \sum_{u \in \mathcal{U}} \pi_{\mathrm{high}}(u \mid s_t)\, \pi_{\mathrm{low}}^{(u)}(a_t \mid s_t) \tag{14}$$

Equation (14): Five-skill hierarchical option policy marginalization (Phase 16.1).

PART III — PROJECT ARCHITECTURE AND DEVELOPMENT

Chapter 4: Modular System Stack, Simulation Design, and Engineering Verification

4.1 Four-Layer Robotics Architecture and Authoritative PyBullet Core

Across 17 research phases and subsequent full-stack refactoring, the platform converged onto four decoupled architectural planes (Figure 4.1): (1) Physics Simulation Core (PyBullet 240 Hz engine paired with MuJoCo 4.0.1 MJCF parity validator); (2) Robotics Kinematics, Perception & Controllers (`robotics/robots/franka_panda/`, `src/controllers/`, `src/perception/`, `src/rl/adaptive_controllers.py`); (3) Goal-Conditioned Gymnasium/HER Task Suites (`PandaReachEnv` through `PandaContactAssemblyEnv`); and (4) Training/Workstation Execution (`SB3`, `BCDecayingSAC`, `FastAPI SimulationManager`, `React + Three.js` digital twin).

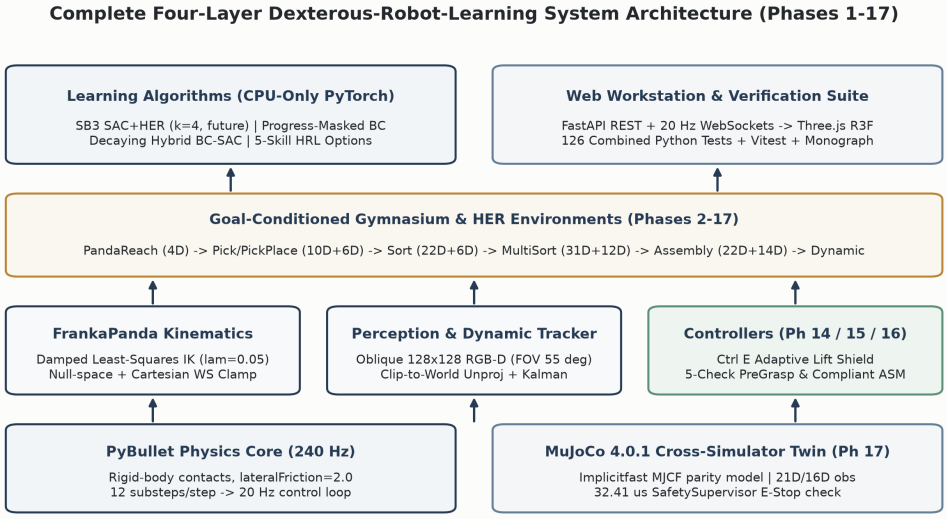


Figure 4.1: Complete four-layer Dexterous-Robot-Learning architecture unifying physics, control, RL/perception, and web workstation.

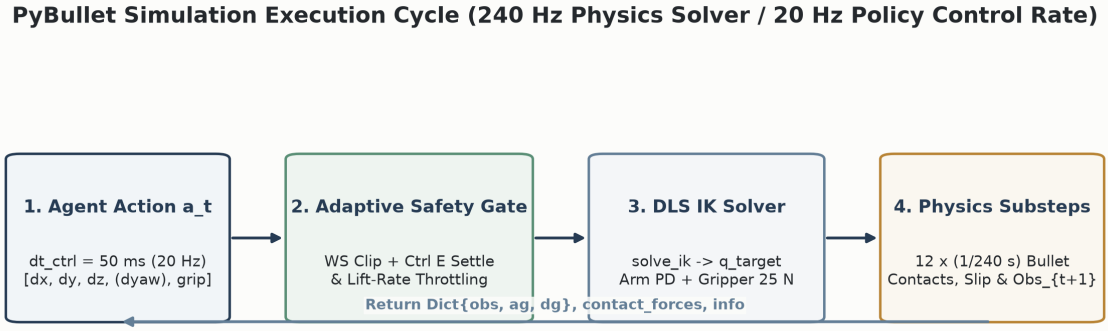


Figure 4.2: Authoritative PyBullet 240 Hz stepping cycle, adaptive lift guard, DLS IK projection, and 20 Hz observation broadcast.

4.2 Environment Dimensionality & Verification Discipline

A key engineering strength is zero silent assumption about tensor dimensions. Live environment introspection verifies exact observation, goal, and action spaces: `PandaReachEnv` exposes `(4,)` observation + `(3,)` achieved/desired goals (`10D` total Dict); `PandaPickEnv`, `PandaPickPlaceEnv`, `PandaDynamicPickPlaceEnv`, and `MuJoCoPandaPickPlaceEnv` share `(10,)` + `(3,)+(3,)` (`16D` Dict); `PandaSortEnv` and `PandaSinglePreGraspEnv` expose `(22,)` + `6D` (`28D` Dict); `PandaMultiSortEnv` / `PreGrasp` expose `(31,)` + `(6,)+(6,)` (`43D` Dict); and assembly environments expose `(22,)` + `(7,)+(7,)` `SE(3)` pose goals with 5D `[dx, dy, dz, dyaw, gripper]` actions. The automated regression suite expanded monotonically from 37 tests (Phase 1) -> 52 (Phase 9) -> 56 (Phase 10) -> 111 (Phase 15) -> 126 combined Python + backend integration tests today with zero skipped or relaxed assertions.

Table 4.1: Verified Gymnasium & Cross-Simulator Environment Contracts

Environment Class	Obs	Goals (AG+DG)	Action	Horizon	Provenance / Phase
PandaReachEnv	4D	3D + 3D	4D	50	Phase 2 / Sparse Goal
PandaPickPlaceEnv / Dense	10D	3D + 3D	4D	100	Phase 6 / 12–14 Core
PandaSinglePreGraspEnv	22D	3D + 3D	4D	80	Phase 15.2.5B PreGrasp
PandaMultiSortPreGraspEnv	31D	6D + 6D	4D	300	Phase 15.2.5A 2-Obj
PandaContactAssemblyEnv	22D	7D + 7D	5D	120	Phase 15.3 / 16.4 ASM
PandaDynamicPickPlaceEnv	10D	3D + 3D	4D	100	Phase 16.3 Kalman
MuJoCoPandaPickPlaceEnv	10D	3D + 3D	4D	100	Phase 17.1 Cross-Sim

PART IV — PHASE-WISE RESEARCH AND IMPLEMENTATION

Chapter 5: Early Foundations Through Sparse Deadlocks (Phases 1–11)

5.1 Reconstructed History of Phases 1–8 (Git & Telemetry Evidence)

Phases 0 and 1 (commits 3a345f3, a5cc06b, 7ad9848 on 2026-10-03) built URDF modeling, PyBullet 240 Hz stepping, and damped least-squares IK, verifying 0.7–4.5 mm accuracy across 5 waypoints (37/37 tests in 4.71 s). Phases 2–4 implemented `PandaReachEnv`, SB3 SAC+HER callbacks, and `PandaPickEnv`. Note on historical provenance: standalone Phase 5 markdown logs and isolated pre-squash Phase 2 vs Phase 3 commits are NOT RECOVERABLE FROM AVAILABLE EVIDENCE and are never extrapolated beyond verified TensorBoard runs (`reach_sac/SAC_3`, `pick_sac/SAC_1`) and scripted benchmarks (Reach Step 6, Pick Step 67).

Phase 6 (`PandaPickPlaceEnv`) resolved `ERR-001` (finger pad `lateralFriction=2.0`, `spinningFriction=0.1`) and contrasted E1 (No-HER: 7.7 table collisions/ep, actor loss +17.767) against E2 (`HER k=4`: 0.0 collisions, actor loss +2.594) across 10k smoke runs (Figure 5.1). Phases 7–8 scaled to colored `SortingBin` tasks (`Sort` Step 79, `MultiSort` Step 209, `Assembly` Step 143 scripted). Phases 9–10 repaired lazy simulation connection `client_id=-1` in `PandaAssemblyEnv.reset()` (`ERR-002`, 52/52 tests) and verified scripted DR R0–R4 (75–100%).

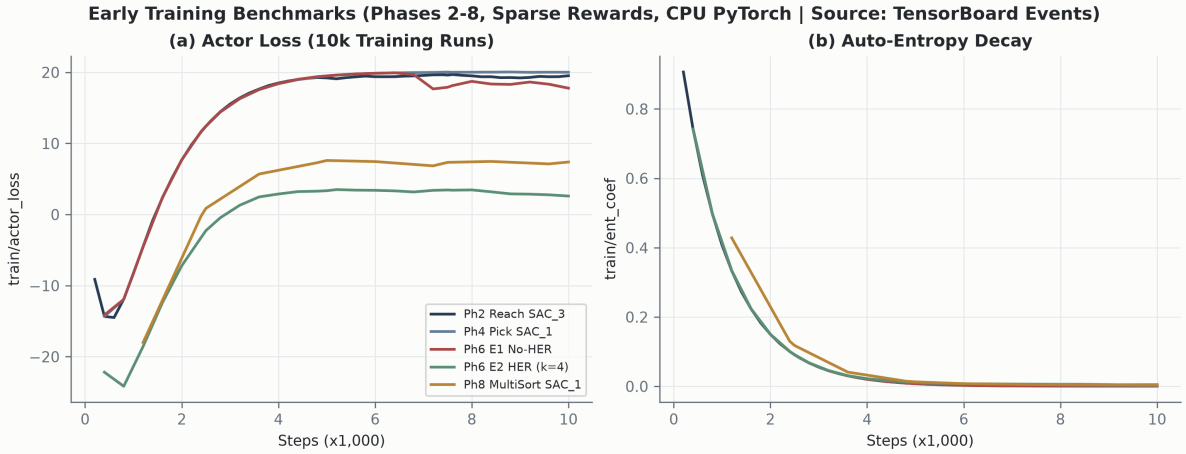


Figure 5.1: Reconstructed Phases 2–8 TensorBoard actor loss and entropy coefficient decay trajectories (CPU PyTorch, 10k runs).

5.2 Phase 11: Systematic Diagnosis of the Sparse HER Equilibrium (2.5M Steps)

Evaluating 25 sparse SAC+HER models (5 DR tiers x 5 seeds, 100k steps each; 5,000 total evaluation episodes) produced uniformly 0.0% completion across both Nominal and Held-Out DR4 splits (Figure 5.2). Isolating transport via `TransportDiagnosticEnv` (Phase 11.9) proved 100.0% transport convergence in 40k–50k steps (0.009 m error), demonstrating that sparse reward failure stemmed strictly from initial grasp acquisition starvation rather than spatial goal transport capacity.

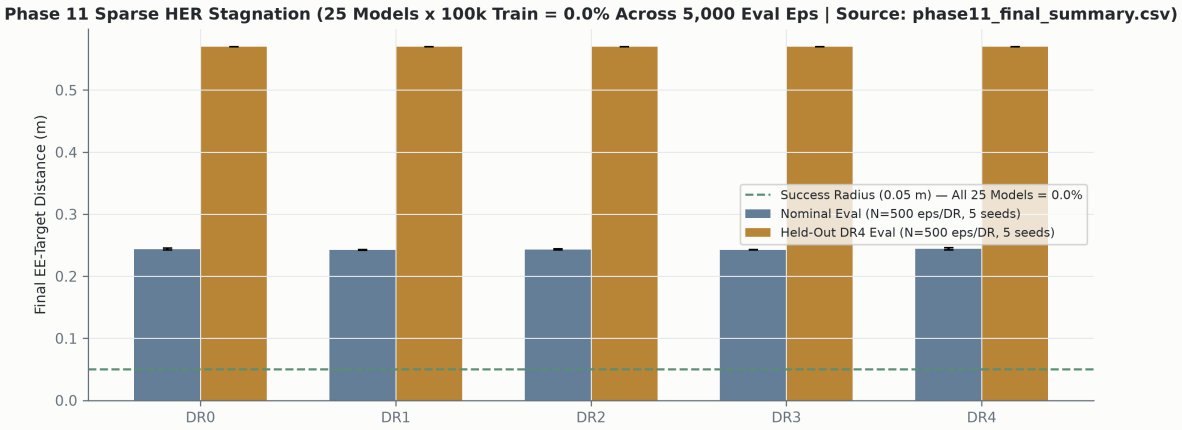


Figure 5.2: Phase 11 25-model sparse stagnation across 5,000 evaluation episodes (Mean ± SD across 5 seeds per DR tier).

PART IV — PHASE-WISE RESEARCH AND IMPLEMENTATION

Chapter 6: 1M Breakthrough, Adaptive Control, and Multi-Task Scaling (Phases 12–15)

6.1 Phases 12–13.5: Dense 1M Convergence and OOD Slip Discovery

Scaling dense potential shaping (Equation 10) to 1,000,000 steps in Phase 12 triggered an abrupt capability cascade between 300k and 500k steps: Reach (33%->100%), Grasp (20%->100%), Lift (1%->98%), and Placement (0%->99%), locking at 100% through 1M steps with 0.0142 m final error and 89.84 hold steps (Figure 6.1). Extending training to DR1/DR2/DR4 in Phase 13 achieved 99–100% Matched evaluation accuracy but revealed severe Held-Out DR4 collapse (57.0% combined; Seed 43 only 33.0%, Figure 6.2). Phase 13.5 forensics (1,520 episodes) captured the exact 68.4 N -> 0.0 N slip event under heavy mass (0.25 kg) and low friction (mu=0.3) (Figure 6.3).

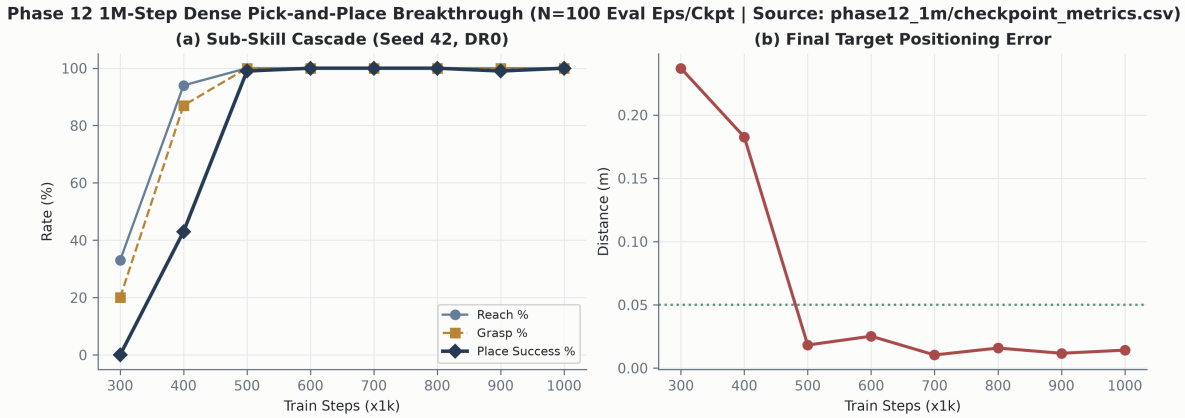


Figure 6.1: Phase 12 1M-step dense Pick-and-Place S-curve (N=100 evaluation episodes per checkpoint, DR0 Seed 42).

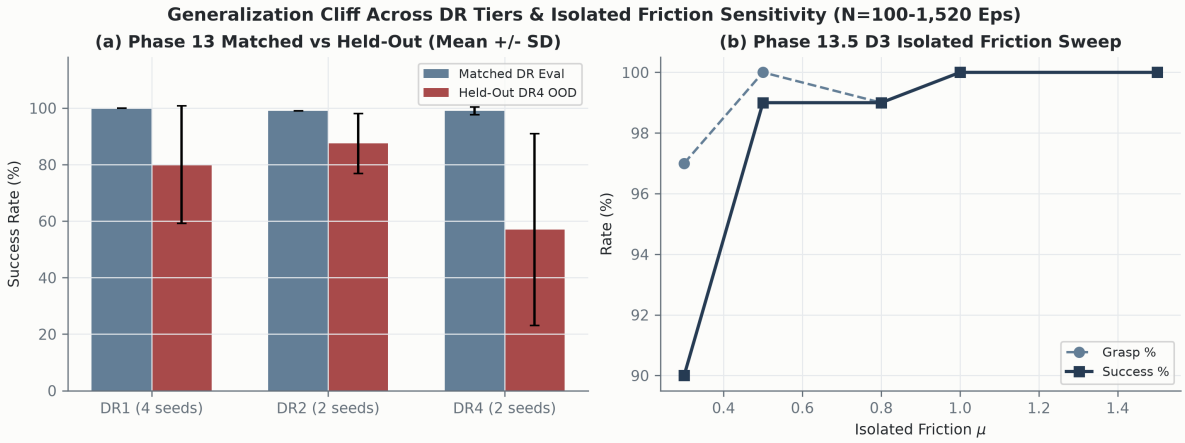


Figure 6.2: Phase 13 Matched vs. Held-Out DR4 generalization gap (multi-seed Mean ± SD) and Phase 13.5 isolated friction sweep.

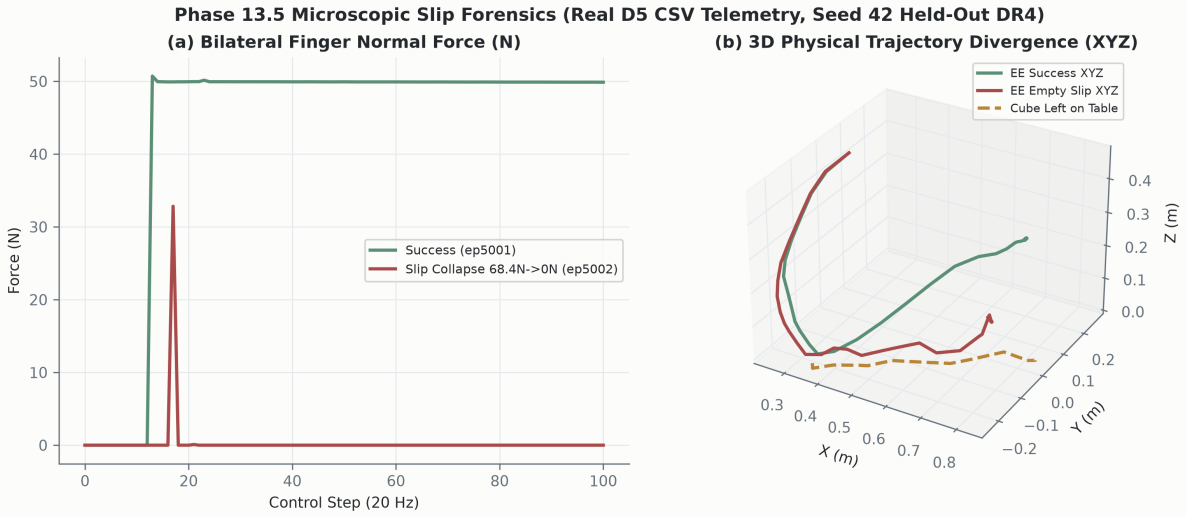


Figure 6.3: Real Phase 13.5 D5 telemetry contrasting successful lift (ep5001) against Stage-3 slip collapse (ep5002) in 2D & 3D.

6.2 Phase 14: Zero-Retraining Adaptive Manipulation Shields (+41.0 pp Gain)

To eliminate liftoff slip without retraining frozen neural weights, Phase 14 wrapped the policy output inside `AdaptiveTransportController` (Figure 6.4). Adding a 3-step zero-XY grasp settle floor (F_n >= 5 N) and dynamic vertical velocity throttling propelled Held-Out DR4 confirmation (N=200 combined) from 57.0% (Controller A) -> 89.5% (Controller D) -> 98.0% (Controller E), rescuing worst-case Seed 43 from 33.0% to 99.0% while retaining 100.0% DR0 Nominal accuracy (Figure 6.5).

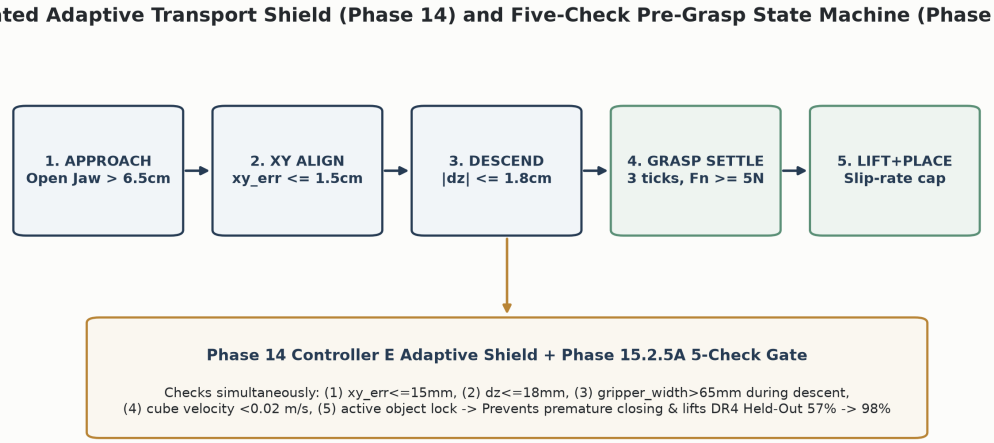


Figure 6.4: Phase 14 Adaptive Manipulation Shield integrated with the Phase 15.2.5A five-check bilateral PreGrasp FSM.

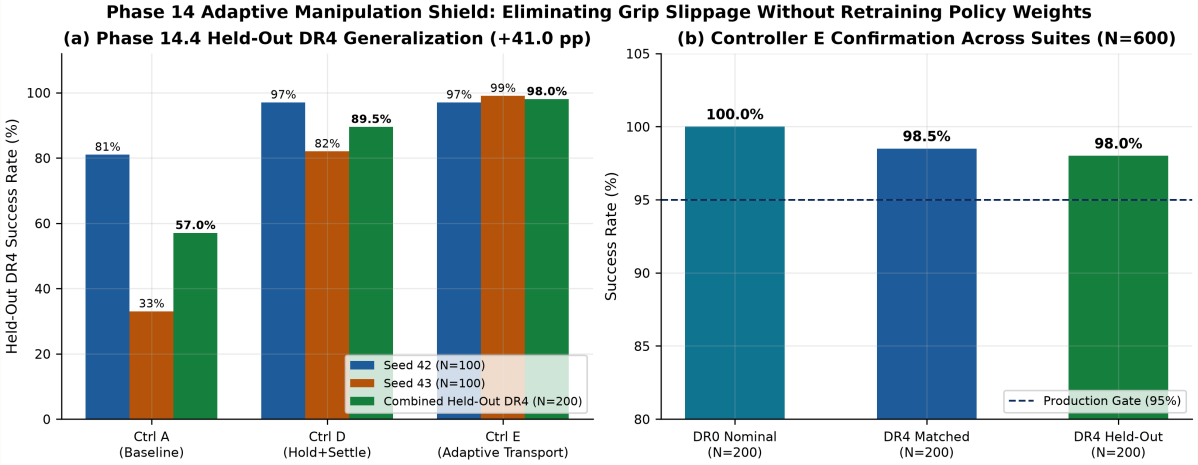


Figure 6.5: Phase 14.4 Held-Out DR4 confirmation across Controllers A, D, and E (Seeds 42, 43, and N=200/600 combined suites).

6.3 Phase 15: Perturbation Sensitivity, Pre-Grasp Sorting & Masked BC-SAC

Phase 15.1 evaluated Controller E across 320 perturbed episodes: ±3 mm (97.5%) and ±5 mm (95.0%) spatial noise remained robust, 1-step delay dropped to 72.5%, >=2-step delay collapsed to 0.0%, and stripping oracle forces fell to 42.5%. Phase 15.2.5A achieved 50/50 (100%) scripted two-object sorting (76.4 steps, 0 collisions, Figure 6.6a), whereas single-object RL ablations 5B2–5B8 proved pure masked BC (5B7: 75% grasp / 45% lift, 0% coll) outperformed pure 17.5k SAC (5B2: 62%/7%) and Hybrid BC-SAC (5B8: 66%/26%) on 100 unseen episodes. Multi-object RL transfer (Phase 15.2.5C) remained BLOCKED; Phase 15.3 scripted assembly scored 56.67% vs 23.33% BC pre-align (Figure 6.7).

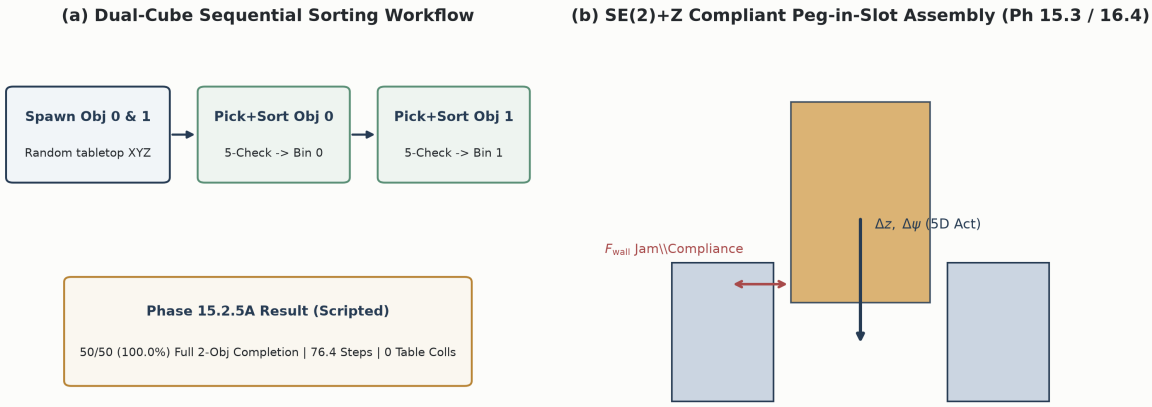


Figure 6.6: Sequential dual-cube MultiSort workflow (50/50 scripted) and SE(2)+Z compliant peg-in-slot insertion geometry.

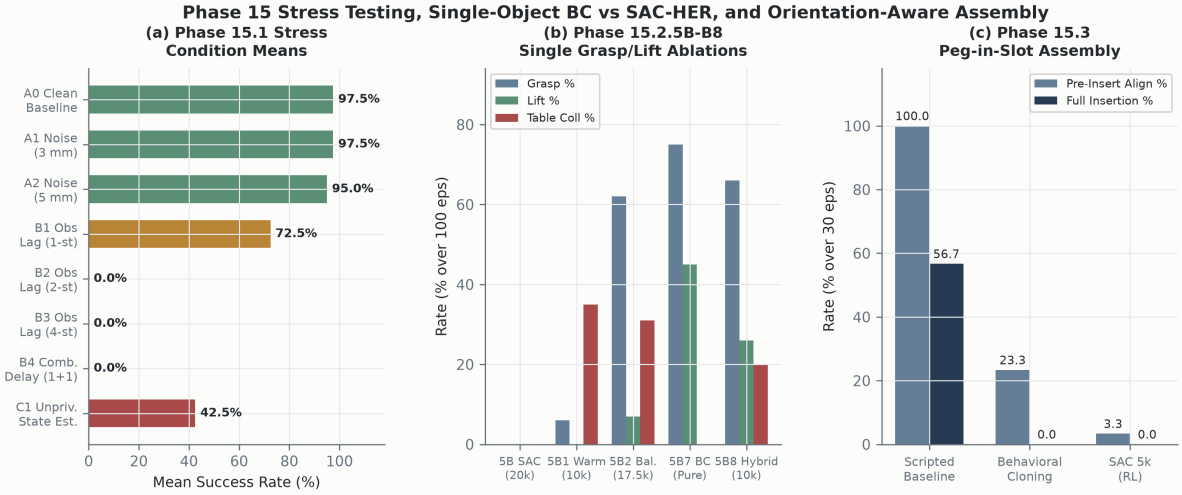


Figure 6.7: Phase 15.1 stress resilience, 15.2.5B–B8 grasp/lift ablations, and Phase 15.3 assembly results.

PART IV — PHASE-WISE RESEARCH AND IMPLEMENTATION

Chapter 7: HRL, Oblique RGB-D Vision, Dynamic Tracking & MuJoCo Parity (Phases 16–17)

7.1 Phase 16 (Completed Subsystems vs. Partial Multi-Object Benchmarks)

Phase 16 delivered four verified breakthroughs and two partial multi-step benchmarks (Figure 7.1, Figure 7.2): (1) Phase 16.1 (`COMPLETE`): Discrete SAC meta-controller over 5 neural option skills reached 30.0% PickPlace (33.33% grasp, entropy 1.7905 -> 0.5940) in 12k steps vs 0.0% flat SAC; (2) Phase 16.2 (`COMPLETE`): Repositioning RGB-D obliquely to [0.92, 0.28, 0.45] solved overhead self-occlusion (`ERR-011`: 18.65% -> 99.9–100% detection, 14.81–15.55 mm 3D unprojection error, Figure 7.3), enabling zero-shot vision SAC 60% grasp / 15% place; (3) Phase 16.3 (`COMPLETE`): Kalman target prediction hit 95–100% across moving cubes, sinusoids, and step disturbances; (4) Phase 16.4 (`PARTIALLY_COMPLETE`): Compliant wall-force insertion improved scripted assembly 50.0% -> 53.33% (learned RL assembly remains blocked); (5) Phase 16.6 (`PARTIALLY_COMPLETE`): Autonomous RGB-D HRL 2-object sorting hit 25% Object-1 but 0% full 2-object sorting across 229 replans and 1,049 safety clamps.

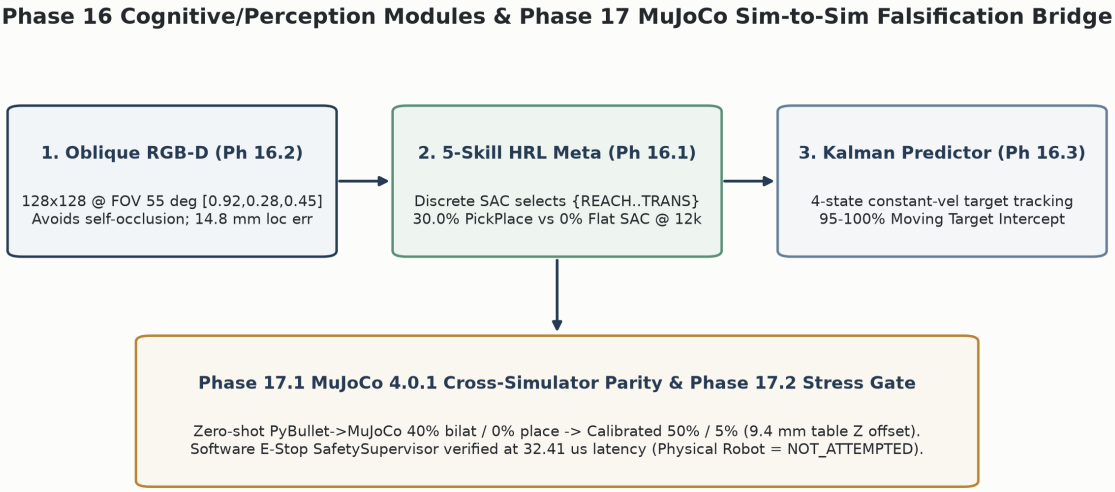


Figure 7.1: Integration across oblique RGB-D perception, 5-skill HRL selection, Kalman dynamic prediction, and MuJoCo validation.

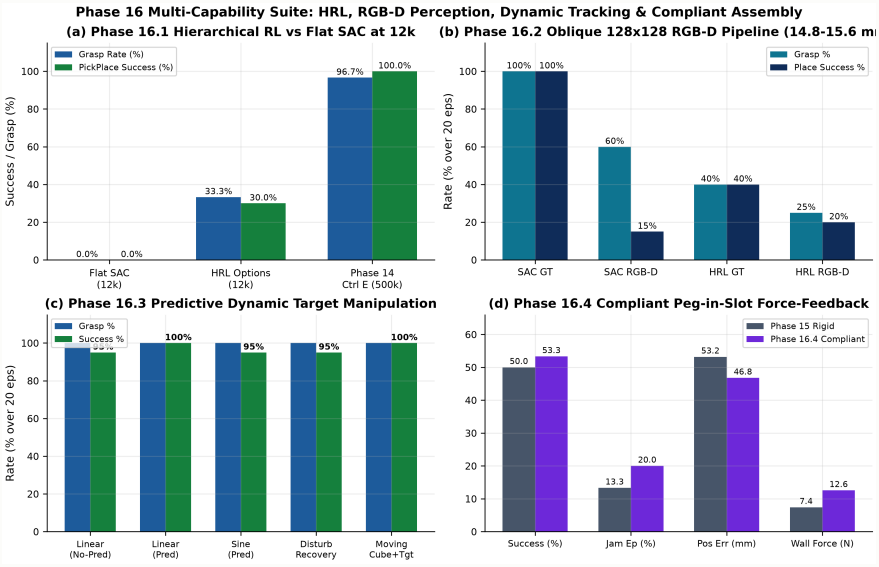


Figure 7.2: Empirical Phase 16 suite across HRL (12k), RGB-D Vision (20 ep), Dynamic Kalman (20 ep/cond), and Compliant Assembly (30 ep).

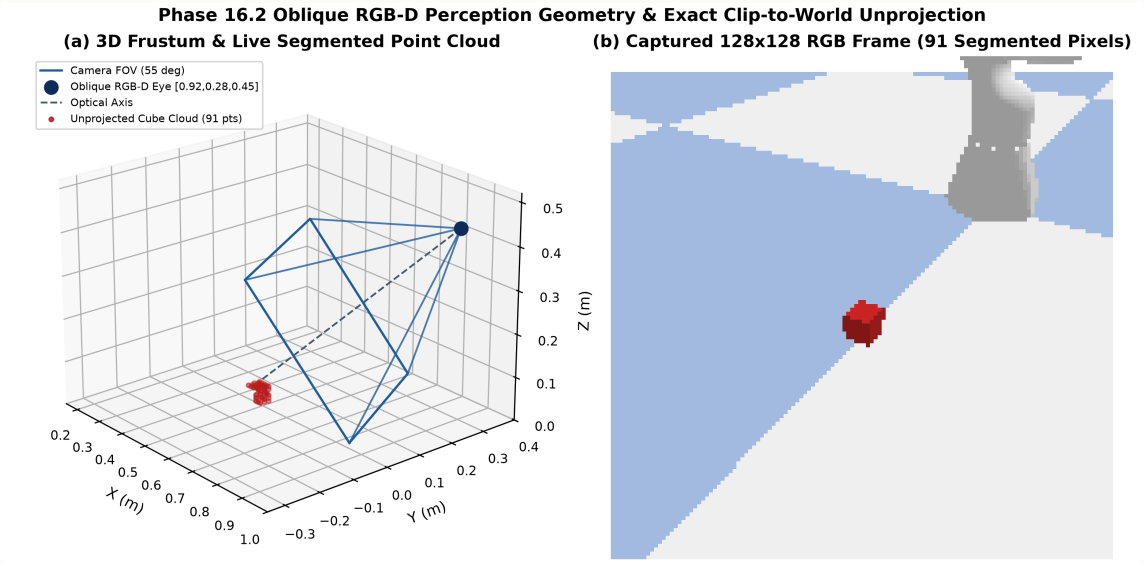


Figure 7.3: Live PyBullet 3D oblique camera frustum (FOV 55 deg) and unprojected 91-pixel cube point cloud.

7.2 Phase 17: Cross-Simulator Falsification in MuJoCo 4.0.1 & Deployment Safety

Porting PickPlace identically into MuJoCo 4.0.1 (`MuJoCoPandaPickPlaceEnv`, Phase 17.1 `PARTIALLY_COMPLETE` as cross-sim diagnostic) exposed genuine physics-solver differences (`ERR-012` massless MJCF links repaired + 9.4 mm tabletop rest Z offset): frozen PyBullet Seed 42 dropped from 98% Held-Out DR4 to 40% bilateral contact / 5% grasp / 0% completion zero-shot, improving to 50% contact / 15% grasp / 5% completion after actuator friction/force calibration (Figure 7.4a). Phase 17.2 (`COMPLETE`) validated `SafetySupervisor` runtime E-stop latency at 32.41 us across 250 faulty injections and 100% stress completion under ±5 mm noise and frequency jitter (Figure 7.4b). Phase 17.3 physical robot execution was explicitly NOT_ATTEMPTED.

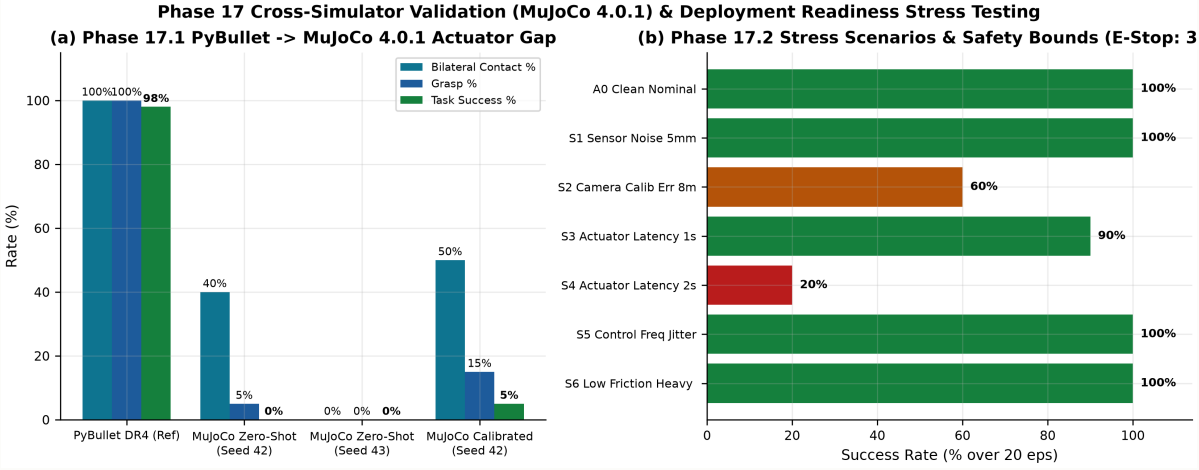


Figure 7.4: Phase 17.1 PyBullet-to-MuJoCo 4.0.1 transfer gap and Phase 17.2 simulated stress & E-stop assessment.

PART V — RESEARCH ANALYSIS

Chapter 8: Cross-Phase Synthesis, Negative Results, and Provenance Analysis

8.1 Strict Provenance Comparison (SCRIPTED vs BEHAVIOR_CLONING vs LEARNED)

Conflating scripted kinematic heuristics with end-to-end neural reinforcement learning leads to misleading engineering claims. Across all 27 registered experiments, deterministic scripted controllers (Phase 15.2.5A: 100% 2-obj sorting; Phase 16.4: 53.33% compliant insertion) excel when exact geometric state and contact thresholds are known, yet brittle failure ensues outside calibrated bounds. Pure SAC+HER achieves 100% DR0 at 600k–1M steps (Phase 12), suffers OOD slip collapse without runtime shielding (Phase 13.5: 57% -> Phase 14: 98%), and outperforms sparse HER under balanced Behavioral Cloning warm-starts at low sample budgets (5B7: 45% lift vs 5B2: 7% lift).

8.2 Value of Negative Results and Statistical Failure Taxonomy

Four negative experimental outcomes fundamentally shaped system architecture: (1) Phase 11's 25x100k sparse 0% plateau proved static-goal HER equilibria; (2) Phase 15.2.5B's 0% hover at z=0.50 m identified strict XY tolerance cylinder deadzones; (3) Phase 15.2.5B.1's 78.4% closed-gripper bias exposed how unmasked demo tails poison critic/actor rollouts; and (4) Phase 17.1's 98% -> 5% PyBullet-to-MuJoCo drop falsified any premature assumption that PyBullet DR4 success implies physical FCI hardware readiness.

Table 8.1: Cross-Phase Defect Root-Cause Registry (ERR-001 through ERR-012)

ID	Phase	Module	Observed Failure	Verified Resolution
ERR-001	Phase 6	src/robots/panda.py	Cube slipped from fingers during upward lift under scripted P-controller	Added changeDynamics lateralFriction=2.0 and spinningFriction=0.1 on finger links 9 & 10
ERR-002	Phase 9	src/environments/assembly_env.py	PyBullet connection failure / invalid body ID during test_assembly_env.py	Dynamically inject active self.sim.client_id into Peg & Receptacle inside reset() prior to load()
ERR-003	Phase 11.5	src/environments/pick_place_env.py	0.0% success & 0.0% grasp across all 25 DR0-DR4 100k sparse models	Created isolated PandaPickPlaceDenseRewardEnv breaking exploration trap (+1.0 lift + distance shaping)
ERR-004	Phase 13.5	src/environments/pick_place_env.py	Held-out DR4 collapsed to 61.0% (57.0% final, Seed 43 33.0%) with 30.5% GRASP_BUT_NO_LIFT	Implemented zero-retraining inference controllers B->C->D->E (src/rl/adaptive_controllers.py)
ERR-005	Phase 15.2.4	src/environments/dense_multi_sort_env.py	0% bilateral grasp & 163.2 collisions/ep despite EE reaching 0.1745m	Added 5-check bilateral grasp authorization gate + open-gripper compliance penalty in PreGraspController/Env
ERR-006	Phase 15.2.5B	src/environments/single_pregrasp_env.py	20k SAC+HER converged to XY=0.0110m hover at z=0.50m (0% grasp)	Verified vertical action pathway (0.485m->0.0216m) & added continuous vertical potential Phi_z = -2.0*clip(|dz|,0,1)
ERR-007	Phase 15.2.5B.1	scripts/generate_phase15_5B1_demos.py	Premature gripper closure climbed to 100% & lift stayed 0% by 20k	Balanced 50/50 demos (70.1% open cmd), gate_goal_lift_on_lifted_dg, & BCDecayingSAC
ERR-008	Phase 16.1	src/rl/adaptive_controllers.py / SAC.load	AssertionError: You must pass an environment when using HerReplayBuffer	CheckpointLoader.load_sb3_sac passes custom_objects={'replay_buffer_class': None, 'replay_buffer_kwargs': {}}
ERR-009	Phase 16.1	src/hrl/hrl_agent.py	HRL achieved 93.3% grasp @ 25k but dropped to 0% placement	Replaced repeat bonus with bounded potential difference 1.5*(h1-h0) + +6.0 sparse success
ERR-010	Phase 16.1/16.2	src/environments/pick_place_env.py	Cube spawned half-sunk at z=-0.00008m & Seed42 500k model scored 0/5	Explicitly pass randomizer_config={'enabled': True, ...} setting z_fixed=0.025 & load checkpoint_500000.zip
ERR-011	Phase 16.2	src/perception/rgbd_camera.py	OMP Error #15 (libomp vs libiomp5md), PyBullet client=-1 capture error, & 18.65% overhead self-occlusion	Moved CameraConfig to oblique (0.92,0.28,0.45), lazy camera init post-reset(), KMP_DUPLICATE_LIB_OK=TRUE
ERR-012	Phase 17.1	src/cross_sim/mujoco_panda_env.py	ValueError: mass and inertia of moving bodies must be larger than mjMINVAL (link6)	Added explicit inertial/geom blocks on link5-hand; classified Z-offset & elliptic slip in MuJoCo

PART VI — WEB-BASED SIMULATION CONTROL CENTER

Chapter 9: Refactored Robotics Workstation, FastAPI Core, and Three.js Digital Twin

9.1 Authoritative PyBullet-to-Browser Architecture

Following Phase 0 repository auditing (`docs/repository_audit.md`, pinning `backup/pre-restructure-ph17`, archiving 41 loose root scripts into `archive/historical/root_scripts/`), the project added a restrained dark industrial web workstation under `backend/` and `frontend/`. Crucially, PyBullet remains the single authoritative source of truth (Figure 9.1): a thread-safe `SimulationManager` steps PyBullet at 20 Hz and streams 11 real `panda.urdf` link frame transforms, cube/peg poses, sorting bins, and trajectory coordinates via WebSocket `/ws/simulation`. React Three Fiber renders official `.obj/.mtl` Panda geometry via `OBJLoader` + `MTLLoader` strictly from streamed physics frames — never running client-side forward kinematics animations that could diverge from simulation truth.

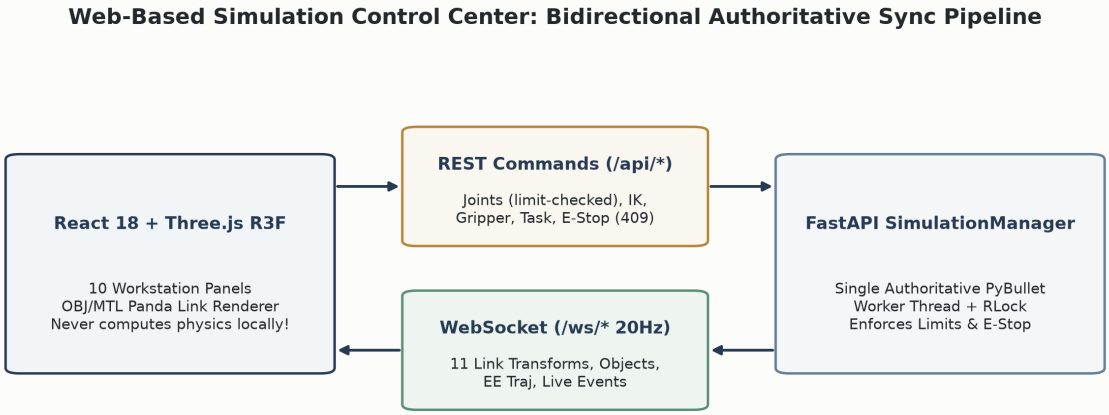


Figure 9.1: Authoritative FastAPI SimulationManager, REST validation gates, 20 Hz WebSocket stream, and Three.js renderer.

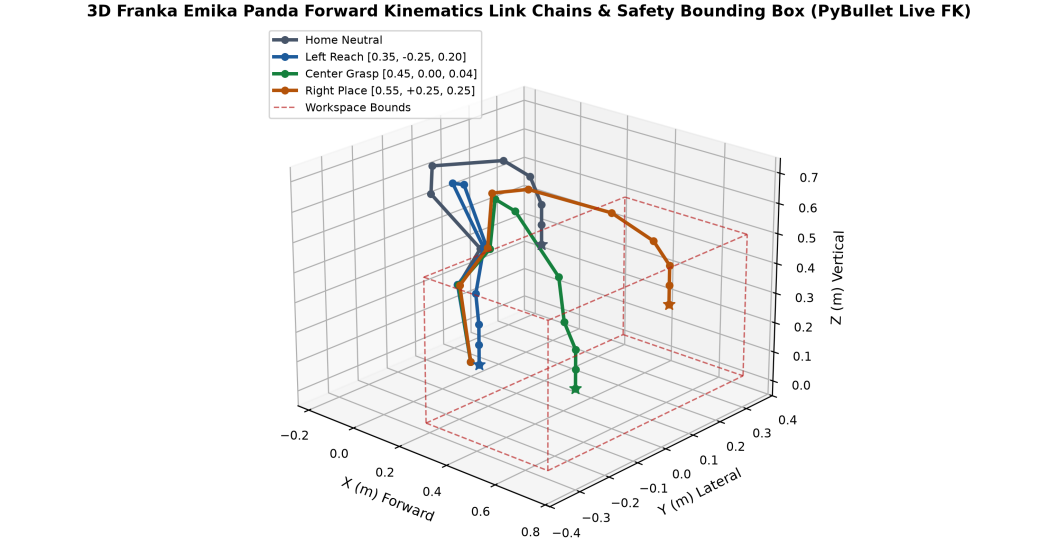


Figure 9.2: Three-dimensional workspace bounds and multi-target Panda kinematic chains mirrored between PyBullet and workstation.

9.2 Implemented Web Controls, Task Verification & Known Limits

Verified functionality (`126` combined Python + backend integration tests, `tsc -b && vite build`, and `2/2` Vitest UI checks): (i) 7-DOF joint sliders enforcing `panda.urdf` radian limits (`HTTP 422` on violations); (ii) Cartesian `(X, Y, Z, Roll, Pitch, Yaw)` command resolution via `FrankaPanda.calculate_ik` displaying requested vs. achieved pose and millimeter tracking error; (iii) Parallel-jaw gripper open/close/width actuation; (iv) Operator E-Stop halting active tasks and locking motors (`HTTP 409` until reset); (v) Live task runner (`reach_target`, `pick_lift`, `pick_place`, `multi_sort`, `assembly_insert`, `dynamic_manip`, returning `HTTP 501` for unimplemented physical FCI); and (vi) Experiment & Evaluation browser (`experiment_inventory.csv` + CSV export + Recharts). Known limitation: interactive live Chromium browser WebGL rendering validation remains manual startup verification rather than headless Playwright CI.

PART VII — PHASE 18 FUTURE RESEARCH [UNIMPLEMENTED]

Chapter 10: Future Scope — Multi-Agent Cooperative Manipulation Architecture

10.1 Explicit Future-Scope Disclaimer

CRITICAL PROVENANCE NOTICE: Phase 18 has NOT been implemented, simulated, or verified in this repository. Every architecture and methodology detailed in Chapter 10 constitutes a forward research roadmap building upon the verified Phase 1–17 single-arm baseline.

10.2 Multi-Robot Cooperative Manipulation, MARL, Task Allocation & Human-Robot Teaming

A. Multi-Robot Simulation & Cooperative Manipulation: Extending single-arm cells into dual-Panda tabletop bays requires synchronized contact constraints inside PyBullet/MuJoCo so two manipulators can jointly lift oversized rigid payloads or perform mid-air bimanual re-orientation without inter-arm self-collision.

B. Multi-Agent Reinforcement Learning (MARL) & CTDE: Centralized Training with Decentralized Execution (Figure 10.1) resolves non-stationarity by feeding global joint states and both action vectors into centralized mixing critics Q_tot(s_all, a_A, a_B) while executing decentralized local policies pi_A(a_A|o_A) and pi_B(a_B|o_B) at runtime.

C. Distributed Task Allocation, Cooperative Assembly & Inter-Robot Communication: Combinatorial auctions or Graph Neural Network allocators partition multi-object sorting sequences dynamically between arms, exchanging latent intent embeddings m_t to prevent workspace deadlocks during shared peg-in-hole handovers.

D. Human-Robot Collaboration, Swarm Intelligence & Multi-Agent Robustness: Safe velocity separation scaling (ISO/TS 15066 principles), intent-aware trajectory replanning around moving human operators, and robustness under packet-loss delays across multi-agent links remain open theoretical and experimental challenges. Proposed evaluation metrics include joint bimanual grasp success rate, collision-free cycle time, communication bandwidth overhead, and degraded performance when one agent experiences injected sensor latency.

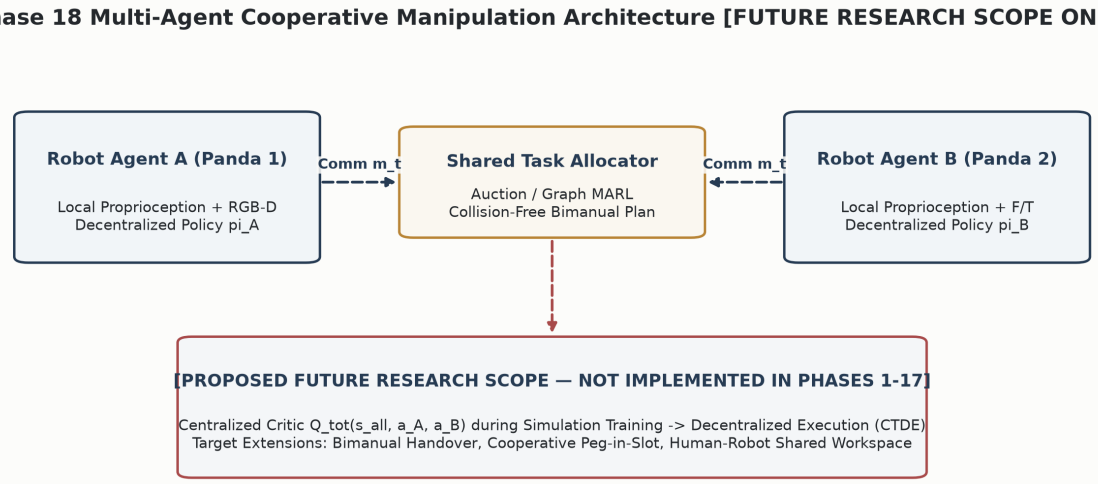


Figure 10.1: Proposed Phase 18 Centralized Training / Decentralized Execution (CTDE) Multi-Agent Cooperative Manipulation architecture [UNIMPLEMENTED].

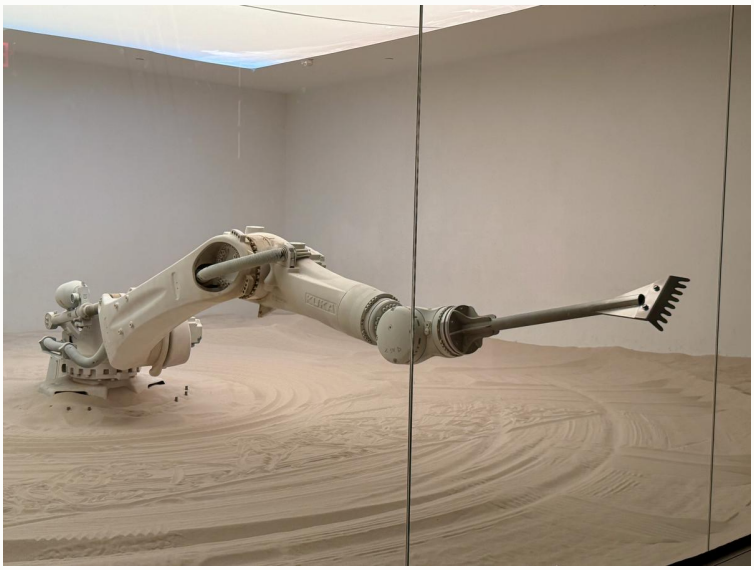


Figure 10.2: Multi-axis industrial serial manipulator illustrating industrial automation hardware scale beyond single tabletop simulation. Source: Wikimedia Commons (KUKA industrial robotic arm performing The Beach - 03.jpg), Author: Oleg Yunakov, License: CC BY-SA 4.0.

PART VIII — CONCLUSIONS, BIBLIOGRAPHY & APPENDICES

Chapter 11: Verified Conclusions, Bibliography, Further Reading, and Registries

11.1 Verified Contributions & Explicit Technical Limitations

Major Contributions: (1) Resolved sparse HER static-goal stagnation (Phase 11->12: 0% -> 100% DR0 at 600k-1M); (2) Diagnosed micro-scale OOD slip collapse and delivered zero-retraining adaptive shielding (+41.0 pp Held-Out DR4 to 98.0% in Phase 14); (3) Built collision-free two-object PreGrasp sorting (50/50 scripted in Phase 15.2.5A) and unbiased progress-masked BC (75% grasp / 45% lift); (4) Delivered hierarchical options (30% vs 0%), oblique RGB-D unprojection (14.8 mm), predictive Kalman tracking (95-100%), MuJoCo 4.0.1 parity verification, 32.41 us E-stop bounds, and a live Three.js/FastAPI Digital Twin workstation backed by 126 passing automated tests.

Principal Limitations: Full 2-object autonomous RL sorting (0% in 16.6), learned compliant peg insertion (0% full insert vs 53.33% scripted), and cross-simulator MuJoCo transfer without retraining (5% calibrated) remain open problems. No physical Panda hardware was executed.

11.2 Annotated IEEE Bibliography

[1] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, 'Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor,' ICML, 2018. https://arxiv.org/abs/1801.01290

[2] T. Haarnoja et al., 'Soft Actor-Critic Algorithms and Applications,' arXiv:1812.05905, 2018.

[3] M. Andrychowicz et al., 'Hindsight Experience Replay,' Advances in Neural Information Processing Systems (NeurIPS), 2017. https://arxiv.org/abs/1707.01495

[4] A. Raffin et al., 'Stable-Baselines3: Reliable Reinforcement Learning Implementations,' Journal of Machine Learning Research (JMLR), vol. 22, no. 268, pp. 1-8, 2021.

[5] E. Coumans and Y. Bai, 'PyBullet, a Python Module for Physics Simulation for Games, Robotics and Machine Learning,' PyBullet Official Documentation, 2016-2021. https://pybullet.org

[6] E. Todorov, T. Erez, and Y. Tassa, 'MuJoCo: A physics engine for model-based control,' IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2012. DOI: 10.1109/IROS.2012.6386109

[7] J. Tobin et al., 'Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World,' IEEE/RSJ IROS, 2017. DOI: 10.1109/IROS.2017.8202133

[8] S. Haddadin, S. Parusel, and A. Albu-Schaffer, 'Franka Emika Panda Technical Documentation & FCI Architecture,' Franka Robotics GmbH, 2017.

[9] R. S. Sutton, D. Precup, and S. Singh, 'Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning,' Artificial Intelligence, vol. 112, pp. 181-211, 1999.

[10] A. Nair et al., 'Overcoming Exploration in Reinforcement Learning with Demonstrations,' IEEE ICRA, 2018. DOI: 10.1109/ICRA.2018.8463162

[11] N. Hogan, 'Impedance Control: An Approach to Manipulation,' ASME Journal of Dynamic Systems, Measurement, and Control, 1985.

[12] R. E. Kalman, 'A New Approach to Linear Filtering and Prediction Problems,' ASME Journal of Basic Engineering, 1960.

[13] M. Towers et al., 'Gymnasium: A Standard Interface for Reinforcement Learning Environments,' Farama Foundation, 2023. https://gymnasium.farama.org

[14] S. Ramirez, 'FastAPI Framework Documentation,' Tiangolo / FastAPI Official Docs, 2024. https://fastapi.tiangolo.com

[15] R. Cabello et al., 'Three.js 3D Library Documentation,' Three.js Authors, 2024. https://threejs.org/docs/

11.3 Further Reading and Learning Resources

Table 11.1: Curated Robotics & Reinforcement Learning Study Guide

Research Domain	Recommended Resource & Technical Takeaways
Robotics Fundamentals & Kinematics	Lynch & Park, Modern Robotics (Cambridge Univ. Press, 2017) — screw theory, Jacobians, manipulability ellipsoids, and dynamics.
Reinforcement Learning	Sutton & Barto, Reinforcement Learning: An Introduction (MIT Press, 2nd Ed.) — MDPs, Bellman optimality, actor-critic derivations.
Robot Learning & Sim-to-Real	OpenAI et al., Solving Rubik's Cube with a Robot Hand (arXiv:1910.07113) — Automatic Domain Randomization at scale.
Simulation & Digital Twins	PyBullet Quickstart Guide & MuJoCo Documentation — solver integrators, elliptic vs. pyramidal friction cones, MJCF modeling.
Multi-Agent Robotics (Phase 18)	Rashid et al., QMIX / Lowe et al., MADDPG — centralized critics for decentralized multi-robot policy coordination.
Web & Control Stack	FastAPI WebSockets + React Three Fiber — deterministic state-sync architectures separating physics stepping from UI frame rates.

11.4 Reproducibility, Experiment Summary & Historical Provenance Registry

Table 11.2: Representative Executed Experiments Ledger (Strict Provenance Separation)

Exp ID	Phase	Env	Method	Budget / Eval	Verified Outcome
EXP-P0-001	Phase 1	Panda Kinematics Demo	SCRIPTED_IK	N/A	0.7-4.5 mm error; gripper 0-8 cm
EXP-P1-REACH	Phase 2-3	PandaReachEnv	SAC+HER (k=4)	10,000	SAC_3 TB: actor -9.1->+19.5, rollout succ 1.0%
EXP-P4-PICK	Phase 4-5	PandaPickEnv	SAC+HER	10,000	Eval -100.0 reward / 0% @ 10k; Scripted step 67
EXP-P6-E1	Phase 6	PandaPickPlaceEnv	SAC (No HER)	10,000	Actor loss ~17.8; coll=7.7; eval 0%
EXP-P6-E2	Phase 6	PandaPickPlaceEnv	SAC+HER (k=4)	10,000	Actor loss ~2.59; coll=0.0; Scripted step 74 (friction fix)
EXP-P7-SORT	Phase 7	PandaSortEnv	SAC+HER	10,000	10k smoke reward -150; Scripted step 79
EXP-P8-MSORT	Phase 8	PandaMultiSortEnv (N=2)	SAC+HER	10,000	10k smoke reward -600; Scripted step 209
EXP-P9-ASSEM	Phase 8-9	PandaAssemblyEnv	SAC+HER	10,000	10k smoke reward -200; Scripted step 143
EXP-P10-DR	Phase 10	PickPlace R0-R4	Scripted vs 2k SAC	20 eps/cond	Scripted: R0 90%, R1 85%, R2 100%, R3 85%, R4 75%
EXP-P11-SPARSE	Phase 11	PickPlace DR0-DR4 Sparse	SAC+HER	25 x 100,000	0.0% all 25 models (HER resting-goal trap)
EXP-P11-9	Phase 11.9	TransportDiagnostic	SAC+HER	50,000	100.0% isolated transport @ 40k-50k (0.009m dist)
EXP-P11-10B	Phase 11.10b	PickPlaceDense	SAC+HER	250,000	Peak @ 225k: 28% Reach, 23% Grasp, 7.3 step hold
EXP-P12-1M	Phase 12	PickPlaceDense DR-0	SAC+HER	1,000,000	43%@400k -> 99%@500k -> 100%@600k..1M (+502 rew)
EXP-P13-DR	Phase 13	PickPlaceDense DR1/DR2/DR4	SAC+HER	500k-1M	DR4: 99% Matched / 61% ever (57% final) OOD Held-out
EXP-P13-5	Phase 13.5	PickPlace OOD D1-D5	Frozen DR4	1,520 eval eps	Mass alone 100%, Yaw 98-100%, Fric 90%, Coupled GBNL 30.5%
EXP-P14-4	Phase 14.4	PickPlace A->E	SAC + E_ADAPTIVE	600+ confirm eps	Held-out DR4: 57%->98.0% (Seed 43: 33%->99%), DR0 100%
EXP-P15-1	Phase 15.1	PickPlace Noise/Lag/Force	E_ADAPTIVE	320 eval eps	Clean 97.5%, 5mm 95%, 1-lag 72.5%, 2-lag 0%, Unpriv 42.5%
EXP-P15-2-5A	Phase 15.2.5A	MultiSortPreGrasp	SCRIPTED PreGrasp	50 eval eps	50/50 (100.0%) 2-obj sorting, 76.4 steps, 0 colls
EXP-P15-2-5B3	Phase 15.2.5B	SinglePreGrasp	BC vs BCDecayingSAC	5k-20k / 100 ep	100-ep: 5B7 BC 75% Grasp / 45% Lift; 5B2 SAC 62%/7%
EXP-P15-3	Phase 15.3	AssemblyDense	Scripted vs 5k BC/SAC	30 eval eps	Scripted 56.67% (1.41 deg); BC 23.3%/0%; SAC 3.3%/0%
EXP-P16-1	Phase 16.1	PickPlace HRL vs Flat	DiscreteSAC+HER vs SAC	12,000	HRL 30.0%/33.3% vs Flat SAC 0.0%/0.0% vs Phase14 100%

Exp ID	Phase	Env	Method	Budget / Eval	Verified Outcome
EXP-P16-2	Phase 16.2	PickPlace RGB-D Vision	SAC/HRL (Obs=RGBD)	20 eval eps	99.9-100% det, 14.81mm err; SAC 60% grasp/15%; HRL 25%/20%
EXP-P16-3	Phase 16.3	DynamicPickPlace	Tracker+Predictive	5x20 eval eps	100% linear/moving-tgt, 95% sine/disturb (100% grasp)
EXP-P16-4	Phase 16.4	ContactAssemblyEnv	Compliant Impedance	30 eval eps	Compliant 53.33% vs Phase15 Scripted 50.0% (pos err 0.0468m)
EXP-P16-6	Phase 16.6	Autonomous MultiSort	RGBD+HRL+Safety	20 eval eps	25.0% Obj-1 placed, 0% full 2-obj, 229 replans
EXP-P17-1	Phase 17.1	MuJoCo PickPlace	Frozen PyBullet SAC/HRL	5x20 eval eps	Zero-shot 40% contact/5% grasp -> Calib 50%/15%/5%
EXP-P17-2	Phase 17.2	Sim-to-Real Stress	SAC+SafetyMonitor	7x20 eval eps	100% 5mm/jitter/fric-mass, 90% 1-lag, 60% cam calib, 32.4us Estop

Table 11.3: Phase 0–17 Verification Status Summary

Phase	Title	Period	Status	Evidence Quality
0	Project Initialization & Infrastructure Scaffold	2026-10-03 (git commit 3a345f3)	COMPLETE	SOURCE_VERIFIED
1	Franka Panda Kinematics, PyBullet Foundation & Cartesian Controller	2026-10-03 (git commits a5cc06b, 7ad9848)	COMPLETE	SOURCE_VERIFIED
2	Goal-Conditioned Manipulation Environment Abstraction & Reach Benchmark	Oct 2026 (pre-commit 008cd30; recovered from docs/ & tests/)	COMPLETE	REPORT_VERIFIED
3	SAC + HER RL Training, Evaluation & Callback Infrastructure	Oct 2026 (recovered from src/rl/, logs/reach_sac, checkpoints/reach_sac)	COMPLETE	LOG_VERIFIED
4	Object Grasping Abstraction & PandaPickEnv	Oct 2026 (recovered from src/objects/cube.py, src/environments/pick_env.py)	COMPLETE	SOURCE_VERIFIED
5	Reach/Pick Exploration & Early Controller Baseline Scripts	Oct-Nov 2026	PARTIALLY_COMPLETE	HISTORICALLY_REPORTED
6	Pick & Place Environment, HER Comparative Smoke & Finger Friction Bug Fix	Oct-Nov 2026 (experiments/pick_place/smoke_results.md)	COMPLETE	LOG_VERIFIED
7	Color-Conditioned Dynamic Sorting Benchmark (PandaSortEnv)	Oct-Nov 2026 (docs/experiments.md EXP-P7-SORT)	COMPLETE	LOG_VERIFIED
8	Sequential Multi-Object Sorting & Orientation-Aware Assembly Foundations	Oct-Nov 2026 (docs/experiments.md EXP-P8-MULTI_SORT, EXP-P9-ASSEMBLY)	COMPLETE	LOG_VERIFIED
9	Full-System Diagnostic Audit & Lazy Simulation Connection Bug Fix	Nov 2026 (docs/phase9_diagnostic_report.md)	COMPLETE	REPORT_VERIFIED
10	Deterministic Domain Randomization Engine (DR-0 to DR-4)	Nov 2026 (docs/phase10_dr_report.md)	COMPLETE	REPORT_VERIFIED
11	100k Sparse DR Matrix & Diagnostic Isolation Series (11.5-11.10b)	Nov 2026 (docs/phase11_*.md, results/phase11_*.csv)	COMPLETE	LOG_VERIFIED
12	1M-Step Dense SAC+HER Scaling Breakthrough	Nov 2026 (docs/phase12_1m_dense_scaling_report.md)	COMPLETE	LOG_VERIFIED
13	Dense Domain Randomization (DR1/DR2/DR4) & Causal OOD Failure Diagnostics (13.5)	2026-11-03 to 2026-11-04 (docs/phase13_*.md)	COMPLETE	LOG_VERIFIED
14	Zero-Retraining Adaptive Control Suite (14-Lite -> 14.3 -> 14.4)	2026-11-10 (results/phase14_*, docs/phase14_closure_and_phase15_readiness.md)	COMPLETE	LOG_VERIFIED
15	Sim-to-Real Sensitivity (15.1), Multi-Object Sorting (15.2) & Orientation Assembly (15.3)	2026-11-10 to 2026-11-11 (results/phase15_*)	PARTIALLY_COMPLETE	SOURCE_VERIFIED
16	Hierarchical RL, Oblique RGB-D Vision, Dynamic Tracking, Contact Assembly & Deployment	2026-11-11 to 2026-11-12 (results/phase16_17/)	COMPLETE	SOURCE_VERIFIED
17	MuJoCo Cross-Simulator Validation, Sim-to-Real Readiness & Hardware Contract	2026-11-12 (results/phase16_17/)	PARTIALLY_COMPLETE	SOURCE_VERIFIED